

[DRAFT] My thoughts on breathing, with an emphasis on dealing with chronic shortness of breath

Issa Rice

2026-01-09

Contents

1	Availability of this document	4
2	Preliminaries	4
2.1	What kind of shortness of breath/air hunger am I talking about?	4
2.2	Why this document?	5
2.3	Warning/disclaimer and disclosures	6
2.4	Ruling out more serious medical problems	7
3	Safety	8
3.1	Accidental meditation	11
3.2	Coughing	12
3.3	Yawning	12
3.4	Breath holds	13
3.5	Tension ping pong/tennis/whackamole	13
3.6	Suffocation feeling	15
4	Different types of shortness of breath	17
4.1	Questionnaire	19
5	The biological basis of shortness of breath	19
5.1	How breathing works	20
5.2	Misconceptions about CO ₂	22

5.3	Chronic overbreathing and the bicarbonate buffer system .	22
5.4	The Henderson–Hasselbalch equation and CO ₂ tolerance .	26
5.5	Upshot	30
6	What does normal breathing feel like? Is healthy breathing the same for everyone?/Is normal breathing universal?	31
7	Can you overshoot your CO₂ levels by doing too much reduced breathing?	32
8	Is Buteyko breathing a cure or does it just mask the root cause?	33
9	A short timeline of my improvements	33
10	Sequence of cumulative constraints of breathing practice	35
10.1	The sequence	36
10.2	Any more stages?	38
11	Clearing your nose	39
11.1	Strategies	41
11.1.1	Nasal wash	41
11.1.2	Mechanical dilation	42
11.1.3	Buteyko-style breathing and nose unblocking exercises	42
11.1.4	Antihistamines and steroids and allergies	43
11.1.5	Other tricks	44
11.2	The better your breathing, the better your nose	45
11.3	Mouth taping and chin strapping	46
11.4	Picking and blowing your nose	46
11.5	Differential diagnosis ideas	47
11.6	What to do when you have a cold (or your body is otherwise producing a lot of mucus)	48
11.7	Breathing trouble when trying to fall asleep	49
12	Allergies	49
13	Techniques vs habits	50
13.1	Buteyko as coping mechanism (i.e. techniques) and Buteyko as habit	51
14	Capnometry biofeedback	51

14.1	The unreasonable effectiveness of capnometers	53
15	Breathing exercises	54
15.1	Main practice: relaxed silent nasal breathing	55
15.1.1	Instructions for the main practice	60
15.1.2	Follow guided audio tracks while doing the main practice	61
15.1.3	Signs you are doing the main practice correctly . .	62
15.1.4	The goal of the main practice is to enter the Good State	63
15.1.5	The goal post-main practice is to maintain the Good State for as long as possible	63
15.1.6	The main practice is surprisingly tricky to do	63
15.1.7	Scheduled practice is necessary: the main practice is distinct from everyday silent nasal breathing	63
15.1.8	Dangers with the main practice (what can go wrong)	64
15.1.9	Results of the main practice depend on both skill and initial state before the session	64
15.1.10	It is possible to spontaneously enter the Good State	65
15.1.11	Tools to help with the main practice	65
15.2	Introductory exercise: Many small breath holds	65
15.3	Intermittent technique: Double inhale technique	66
15.4	Intermittent technique: Slowing the first part of the exhale .	66
15.5	Intermittent technique: watching automatic breaths	66
15.6	Temporary exercise: soft smooth slow long inhales	67
15.7	Temporary exercise: soft long exhales	67
15.8	Intermittent exercise: observe your breathing after waking up	67
15.9	Intermittent technique: distraction	68
15.10	Intermittent exercise: timing how long it takes between big breaths	69
15.11	Intermittent exercise: count how many times you blow your nose per day	72
16	Posture	72
17	Breathing and sleep	73
18	Talking out loud	73

19 Muscle tension	75
19.1 Greedy vs non-greedy relaxation	75
20 Breathing synthesis	76
21 Misconceptions	76
22 Cheat sheet: emergency things to do if you're really having trouble breathing	77
Acknowledgments	79
References	80

1 Availability of this document

This document is available in a few different formats:

- As a webpage, at <https://riceissa.github.io/breathing/>
- As a PDF document, at <https://riceissa.github.io/breathing/breathing.pdf>
- As an ebook, at <https://riceissa.github.io/breathing/breathing.epub>

Additionally, the source markup and commands used to generate all three formats above are available at <https://github.com/riceissa/breathing/>.

2 Preliminaries

maybe a better title is: overcoming chronic shortness of breath by fixing your breathing

2.1 What kind of shortness of breath/air hunger am I talking about?

- feeling like you can't get a good breath in
- trying to take forceful breath by e.g. yawning

- some breaths “feel good”, other ones (even forceful ones) “feel bad” – there’s a binary classification
- no obvious cause: not tied to exercise, seasons, allergen sources, etc.

2.2 Why this document?

there has been a lot written about breathing, both in the academic literature and in self-help/meditation/mindfulness contexts. why am i adding my own words to this vast literature? what value am i adding?

I am writing basically the document I wish someone had handed to me over a decade ago when I first started to experience shortness of breath.

There’s a lot written about air hunger already. the problem is that each source only seems to have a part of the solution. there’s no one-stop place to get everything i need. so in some sense i’m just synthesizing and collecting everything into one place.

but also, i’ve noticed a lot of sources making mistakes or not fully explaining things. so i’ve had to think about some stuff and work it out on my own or with friends, and also by experimenting on my own.

i will be critiquing a lot of sources as i discuss them. i don’t want to sound too negative – it’s only because of all these sources that i’ve made significant progress on overcoming my own breathing problem. but seeing how things are explained incorrectly/what can go wrong is one of the best ways to understand why something is the way it is. see here for more on this pedagogical perspective.

note to self: this document should have a tightly focused scope. it’s to explain what is going on biologically in air hunger, and then explain *how to get out of that state*. all my other random thoughts about breathing should be put somewhere else. ask: will this directly help someone who is suffering from shortness of breath?

a note for breathing experts: if you’re wondering what the philosophy of this document is in a few words, check out the Breathing synthesis section. In short, my philosophy is the “(very gentle) Buteyko method” + the “Alexander Technique” + capnometry biofeedback, with the first two terms in quotes because I’m sort of doing my own bastardized versions of them, and wouldn’t want to claim I am propagating the “real thing”. If you don’t know what any of the three terms mean, don’t worry! They will all be explained in the course of this document.

i think a lot of resources are needlessly general, to the point where

the reader has to do a lot of work to figure out which parts to ignore and which parts to follow. trying to satisfy everyone by writing super generally means everyone gets a decent but not great experience. so i'm solving this problem by writing for an extremely specific audience: people like my past self. this will make the document useless or only tangentially relevant to a lot of people, but for the intended audience, it will be *exactly* what they need. i'll try to make this document a self-contained do-it-yourself guide on how to overcome shortness of breath. many other guides don't try to be self-contained; they link out to other things that you can try, or otherwise make you hunt down references or jump through hoops to get them. a printed version of this document, without any internet connection, hopefully with access to a capnometer, and about 2 months (half that if you can already breathe through your nose) should suffice to fix air hunger.

my intention is also not to turn you into some sort of superhuman breathing guru, or cure all your health problems, or whatever. there's only one goal, and it's to eliminate air hunger. i think, if you feel inspired by this document and want to go beyond by practicing more advanced buteyko exercises using other sources, that's a fine thing to do! but this document is limited in scope. it only has the modest goal of fixing the air hunger/shortness of breath and doing so in the safest manner possible.

a lot of works are also wordy and repetitive. this document respects the intelligence of the reader. it emphasizes certain parts to guide the reader, but it will not keep repeating the same things in different words over and over. it will point out common misunderstandings or common misinterpretations, but it will not spend pages upon pages belaboring the point.

Another feature of this document is that everything I talk about in here has been verified to work by at least one person (me); I'm not just coming up with some theory of breathing and then telling people it must work. I am figuring things out by a combination of pre-existing theory and a lot of trial-and-error to see what works best for me.

2.3 Warning/disclaimer and disclosures

THIS IS A DRAFT. THIS IS NOT PERSONALIZED MEDICAL ADVICE. USE AT YOUR OWN RISK. See the section Safety below for more details. (At the same time, I *am* interested in improving this document, so if you think I am recommending something unsafe or I am making some other

sort of error, I'd like to know about it. Email me at riceissa@gmail.com.)

As for disclosures, I have nothing to disclose: I am not selling anything. This document is freely available. I am not affiliated with any of the products or services I recommend in this document. I don't use affiliate links.

also, while i've made significant progress on overcoming shortness of breath, as of this writing i *still experience it daily*. so i think there's still insights i need to discover. but i think i am on the right track in a way that i was *not* before mid-2023, so i think at this point it makes sense to start writing things up and collecting it all in one place, so that as i keep learning i can edit this document. ultimate goal of this document: hopefully in a few months my breathing will be fixed forever and this document will also completely reflect my understanding of how i fixed it.

so yeah, i want to be honest that i haven't figured it all out; i won't pretend i've figured it all out while leaving readers in the dark feeling confused whether they're missing something.

i probably said this somewhere else in the document already, but for now this document is more of a research journal where i come to write down my new ideas about breathing. i think it *can* be used as instructions to replicate what i did to improve my breathing, but it's not mainly intended to be that, *at least for now*. In the future, I want to turn this document more into a DIY guide to fix breathing problems, but first i need to make sure my ideas are correct and can *fully* resolve my own breathing problems. and then i need to make sure the techniques are general enough to work for others with similar problems.

2.4 Ruling out more serious medical problems

If you have the same kind of shortness of breath that I have, then you *do* have a medical problem (this is defended at length in the section The biological basis of shortness of breath), but first, it's good to rule out other more serious medical problems, or medical problems that mainstream medicine knows how to treat.

I think as a first step, it's a good idea to rule out any standard problems with the lungs or other body organs, by e.g., doing a chest x-ray, pulmonary function test, blood test, laryngoscopy to rule out vocal cord dysfunction, etc. I will say more about what tests I got done and why. The breathing stuff I describe later in the document *might* help with your breathing anyway,

even if you have a problem with your lungs, but you definitely want to know if you do have problems and to get proper treatment for those before progressing to any breathing modification.

<https://youtu.be/G3YHBYVxmI?t=1049> – Chris Gilbert lists a bunch of possible reasons for hyperventilation that *aren't* just anxiety/stress.

basically, get yourself checked out by one or more doctors, go through the standard protocol, and if the doctors tell you that there's nothing wrong or that your breathing problems are just due to anxiety, *then* come back to this document and try using the methods in it. the only exception i'm aware of so far is asthma. if you have asthma, then supposedly buteyko breathing is also effective, so you can get *both* mainstream medical treatment (e.g., inhalers) and try the things in this document (though i don't have any first- or second-hand experience here so you should probably just go find a buteyko book and follow that instead).

some other possible things that could be going on, where your CO₂ is low but it's not because of a dysfunctional breathing habit: COPD (alveolar and arterial CO₂ mismatch), metabolic acidosis (ketoacidosis, lactic acidosis) (Litchfield and Reamer 2023, p. 29).

see also Box 1.1 ("When is overbreathing acceptable?") in Chaitow, Bradley, and Chris Gilbert (2014, p. 6)

[TODO: add more details]

3 Safety

I want to make two seemingly contradictory remarks about the safety of trying various breathing techniques.

1. Because I am *already* experiencing significant shortness of breath, just by breathing the way I already am, I have a greater appetite/tolerance for air hunger and unsafe breathing. Whether I like it or not, I am already experiencing some significant amount of risk just by existing in my current way.
2. I have the impression that a lot of breathing coaches and such don't emphasize safety enough. One of the reasons I didn't get into Buteyko breathing earlier (despite having known about it since early 2022 or so) is that I was put off by how dangerous it seemed.

When I started practicing silent nasal breathing, I got a few weird symptoms including head pressure, throat tightening (it felt as if I was being subtly choked), and possibly others. (Since I have a mysterious chronic illness, I have a daily occurrence of weird physical symptoms, so I am already quite used to dealing with random new symptoms, and it's often hard to pin down *why* a new symptom started happening, but in the case of head pressure and throat tightening, I am pretty confident it happens because of the breathing.)

The other thing I want to say about safety. It's not helpful when people just tell you a blanket "if you have health issues, don't attempt this". Part of *why* I'm interested in fixing my breathing is because I already have breathing and other health issues. I shouldn't automatically be turned away from trying to solve these issues! So, I don't want to *ignore* safety issues, but I also don't want to give useless cowardly ass-covering safety warnings. How to avoid both problems?

i also don't want to just put a big safety warning at the start, and then not mention safety at all later on. safety will be a concern throughout. but, also, i will have a safety section here to give an overview of what you might experience.

because i have a lot of health anxiety i am pretty nervous to try new things. i think a lot of buteyko people don't have this same orientation. like, they are in it to get superpowers or something (like sleeping only 2 hours a day) and not out of a sense of desperation to breathe like a normal person. so in some sense this document is like a maximally cautious, "only in it for the health improvements" version.

Andrey Novozhilov (2012) mentions cleansing reactions that can happen, in the section called "Cleansing reactions". Unfortunately it just lists a bunch of pretty nonspecific symptoms without any guidance on what to do about them. . .

Andrey Novozhilov (2012)'s main suggestion for dealing with problems seems to be "get an expert to coach you". i mean, sure, that's always one solution. but what i'm trying to do here is to write everything down so that one can do all the steps on one's own, as safely and effectively as possible.

buteyko people also mention this, but increased sensitivity to medications can be a thing. i used to (2024-01-07 was last time i took it before breathing changed) be able to take 5–10 drops of methyl factors and it would just give me a bit of energy, but now (2024-01-27) even 1 drop of it seems to send me into irritable/uncomfortable/anxious state. i

haven't noticed this kind of heightened sensitivity with any of my other meds/supplements so far though. because the methyl factors thing has only happened once so far (i'm in the middle of the episode as i write this), it *could* be a one-time fluke, who knows.

i tentatively recommend going pretty slowly (need a better descriptor here...) so that you get adjusted to the changes. i'm not sure how to go slowly on some of the steps (e.g., when i switched from loud nasal to silent nasal, the loud nasal just felt very unnatural so i couldn't go back and titrate it).

if you are already ill in various ways like i am, then maybe it's not a big deal to suffer for a while and make rapid progress. the problem is if it's not *just* being uncomfortable for a while, but if it becomes permanent or if it's a sign of actually dangerous stuff happening. i don't know how to distinguish what each symptom means. hence the recommendation to go slowly.

if done correctly, i think the buteyko relaxed breathing stuff is a lot safer than relying on your own broken instincts about how hard you should breathe. it's only been 2 weeks now since my "breakthrough" on January 15, but i'm noticing a lot less struggle with my breathing, like even the couple of suffocation-like episodes seem milder than some of the stuff i was already experiencing before.

notes from roam:

- one of the things i don't like about buteyko ppl and also litchfield is that they don't emphasize safety enough... and like what do you *do* if you start to get intracranial pressure or throat tightening or other weird symptoms?
- this is esp a problem in my case b/c i have other health conditions, and buteyko ppl just do a blanket "if you have health problems, don't do buteyko". that's not helpful.
- and this is esp a problem if like me, you start to just automatically make better breathing a habit – then it's hard to STOP doing it. so what do you do then?

update (2024-02-04): 3 weeks into mostly silent breathing, i am having prolonged (what seems like) electrolyte issues. i can't say for sure that this is from the breathing, as i did eat a quite salty takeout meal (and i've always been sensitive to salty foods), but this is the first time it's lasted 72 hours now (in the past similar things would last at most a few hours).

2024-03-14: the throat choking and tense diaphragm thing has been coming back, especially strongly when i talk for a long time (about an hour). but often even when i haven't been talking much. it's the side of the throat, not the center. i haven't been very successful yet figuring out what to do to fix this. i think for me the problem is the inhale, since exhalation is just relaxing all the muscles to let air out. i'm trying to "breathe in" to different parts of the body, like diaphragm, back of throat, nostrils, etc. also trying to more artificially move diaphragm instead of trying to breathe in with my throat muscles, and then gradually toning down the amount of force i use with the diaphragm while still keeping the throat inactive.

TODO: Artour Rakhimov has a page on focal infections (<https://www.normalbreathing.com/buteyko-focal-infections/>), which talks about why you shouldn't do Buteyko, or why you might not get the same progress as a healthy person. Unfortunately for me, probably some of the stuff on the page applies to me (e.g. gut dysbiosis/candida isn't specifically mentioned, but similar enough stuff *is* mentioned that it probably applies to me). A lot of people now with long-COVID (or like me, some other post-acute infection/exposure syndromes) probably have both shortness of breath and some sort of other stuff going on, such that the focal infection stuff applies. At the moment, I'm not sure how to make sense of all of this. I know that doing *some* light Buteyko-esque stuff seems to be helpful at least for me. I'm going very very slowly, not on any kind of schedule, just gradually doing stuff here and there, backing off whenever I have some kind of bad reaction (e.g. see Suffocation feeling below for an example of how I messed up and harmed myself by practicing Buteyko stuff), and this seems like a good strategy for people in similar situations. I want to think more about this, including reading the Artour page in more detail.

3.1 Accidental meditation

as you try to normalize your breath, you're constantly paying attention to the way you breathe, and that's essentially the mental motion that meditators make (at least, when they do the kind of meditation where they focus on their breath). could you be accidentally meditating by trying to fix your air hunger? is that why the intracranial pressure happens? how can we correct the breath without risking accidentally meditating? (most people might have the sense that meditation is harmless, but my current belief is that if you do more than like a few minutes a day, then

meditation requires a lot of care to do correctly, and you can really risk hurting yourself/disrupting your whole life. see stuff like “dark night of the soul”. as they say about meditation, “Better not to start. Once started, better to finish.” you really don’t want to accidentally meditate.)

i don’t have any good answers yet. i’m still working through this myself. for the head pressure, it was somewhat bad for one day, but then after that it just got better on its own, and now i don’t notice it at all. i’ve also been just trying to do other stuff, so that even as i do inevitably bring my attention back to my breath to see if i’m still doing silent nasal breathing, i’m not just sitting there doing that the entire day.

Andrey Novozhilov (2012) even says “Exercise 1 is really another version of meditation using the focus on breathing and relaxation of the breathing muscles as the means for avoiding wandering thoughts.”

as of 2024-02-17, i am less concerned about this. i basically don’t notice any changes in my psychological state after focus on my breathing so much. i think as long as you’re doing other stuff in the day, and not just sitting there and thinking about breathing the whole day for hours at a time, you should be fine.

3.2 Coughing

Buteyko people seem too anti-coughing to me. My uninformed guess is that it’s good to cough sometimes, e.g., to clear mucus or during a viral infection. Then there are other more mysterious forms of cough (e.g., twice in my life for like a month each, I had a habit of little micro-coughs/chest itchy feeling) where I think suppressing the cough is more justified (the second time it happened, doing soft breathing to make the cough not trigger is what finally made it go away).

Andrey Novozhilov (2012) says to cough without opening your mouth, but based on personal experience I think this is a bad idea. There can be too much air coming through your nose which can damage the nose tissue. Anyway, this was never a huge issue for me but people with actual asthma might have to adopt different strategies.

3.3 Yawning

I’ve seen quite often the advice to suppress yawns, but interestingly even the buteyko people say (naturally occurring) yawning is ok. (Yakovleva,

Buteyko, and A.E. Novozhilov 2016, p. 254) I actually agree with this.

As far as I can make out, one of the reasons why natural yawns are okay is that there is naturally a pause after the yawn. So even though it's technically a "mouth breath", it probably doesn't reduce CO₂ levels as much as a normal mouth breath. (I haven't actually experimentally verified this using a capnometer.)

3.4 Breath holds

breath holds in general scare me, and i think it's fine if you don't want to do them. really tiny ones (like 1–5 seconds) might be good, just to get yourself used to the idea that not breathing for a few seconds isn't going to kill you (you can even have a pulse oximeter on your finger as you do breath holds, to prove to yourself that your oxygen saturation doesn't drop or only drops an insignificant amount, like 1 percentage point); so i still do include Introductory exercise: Many small breath holds in my suggested exercises. some buteyko books like artour's have more safety information about breath holds.

this is also one of the reasons i prefer Capnometry biofeedback, because it gives you a way to measure progress without taking Control Pauses. (at least initially, until your baseline CO₂ levels reach a normal 40 mmHg.)

3.5 Tension ping pong/tennis/whackamole

2024-06-05: i seem to be running into a problem where when i try to do buteyko style breathing, or even just silent nasal breathing, i get tension in my diaphragm and/or throat, and if i try really hard to make the tension disappear from one of them (by relaxing that part), the tension moves to the other part. in this way, the tension just keeps bouncing back and forth, never really entering a state where i have no tension. maybe i need to also start humming or something, not just doing relaxed/reduced breathing.

with the throat tension, this may also have something to do with my dysphagia.

post I made on a forum:

I've had what seems like a related issue, where as I have tried to soften/regulate my breathing, my diaphragm region as well as throat get very tense. Sometimes, when I try to work on tension in one part (e.g. the diaphragm), I can get it to loosen but then

the tension moves to another part (e.g. the throat) instead! I've been calling this "tension whackamole" in my head.

What I've been trying is to "relax even better" by playing around with posture (I find it hard to do constructive rest so I play around by sitting up at varying levels of straightness, by standing, pacing around, etc.), focusing on the tense part and trying to relax it (e.g. one thing you could do is to artificially try to tense it up and then relax it, so that you can feel the difference; this is a technique from a common anxiety exercise called progressive muscle relaxation), trying to "redistribute" the effort of breathing to other parts of the body to make sure I'm not artificially restricting any part of my body (e.g. asking if I can move more of the effort to my diaphragm, or upper chest, or throat), visualizing by imagining my inhale going to various parts of the body (e.g. back of throat, upper chest, diaphragm), and trying to really let my body breathe according to the timing it wants rather than trying to pre-empt the inhale. I've found it helpful to follow breathing relaxation-focused audio recordings.

So my guess (although it's just a guess! – it's hard to tell what's going on without being there, more back-and-forth, etc.) would be that you are letting go of your usual tension, but then you are tensing up in a different way (which is noticeable to you possibly because you're using new muscle groups that you don't normally use much when breathing), so as to "get enough air" or because you are artificially restricting the flow of air.

2024-07-29: pressing down on "L1 acupressure points" (right under the collarbone area on each side) and "breathing into" those points, one at a time, *seems* to help a lot with throat tension in particular. But only if I've fixed the upstream cause of it first (usually, the tension comes on because my nostrils are somewhat clogged, and I'm taking more forceful breaths to compensate, and this leads to throat tension; so in this case, I first need to unclog my nostrils). I heard about this idea from KC. I need to play around with this some more before I know for sure that it works.

3.6 Suffocation feeling

If your breathing/nervous system/whatever is sufficiently compromised, and you don't know what you're doing as you practice breathing exercises, scary things can happen to you. The following is an example of this happening to me. (I am only motivated to keep pushing ahead despite such scary experiences because for me the alternative is to experience shortness of breath every day as well as similar scary experiences from talking too much. And because I see *clear, immediate* benefits from doing the exercises in the right ways.)

Scary experience on the night of 2024-06-07/morning of 2024-06-08: On this night I was not able to do my new Buteyko routine around 9:30 p.m., as things were hectic, so I thought I was being smart by trying to do Buteyko (unguided by a recording) while trying to fall asleep in bed. It's supposed to regulate my breathing, and relax my body, so why not, right? But as I did that I started to get the suffocation feeling (that I sometimes get as I'm falling asleep; see salamanderz1 (2023) for a report of what this feels like) and actually had to sit up and gasp for air. So at that point I gave up on doing Buteyko and just went to sleep. I fell asleep fine. But about 1.5 hours later, I woke up feeling short of breath and noticed I was really straining multiple times to get a good breath of air, and failing to get a good one. (Some of the straining was mostly unconscious, not the conscious kind that I normally do while awake, while others were consciously executed.) I checked on my pulse oximeter and it was 98% (pulse was like 65bpm). It's a very specific kind of air hunger, different from turbinate swelling/mucus clogage and also different from the diaphragm tension stuff I've been having recently.

As I was lying on my belly trying to figure out what to do, watery mucus ran down my nose so I blew my nose, and I just naturally yawned a few times, which made the air hunger better. I then tried going back to sleep, but failed as the air hunger came back.

So at this point, I tried watching the experience more to see what was going on. I noticed that the suffocation feeling happens on the inhale (rather than on the exhale or during the pause after the exhale, which is when a lot of other shortnesses of breath tend to happen). It was as if somehow I was accidentally pre-empting the inhale rather than letting the reflex do the work, and when that's the case it somehow feels bad. Sometimes even during the inhale I started to feel the panic of suffocation. It felt like I had

“forgotten how to breathe”, but in a very specific way.

I also noticed that as I was lying in the dark, if I was tapping on my arm during the inhale I could breathe fine. But if I stopped, my brain would freak out when the inhale was about to happen. And even more oddly, it felt like different lying positions would affect my breathing differently: if I lay down in this weird diagonal way on my back (where mostly the right side of my back is on the mattress, while the left side is more in the air), my breathing seemed to normalize, whereas if I tried curling up on my left side, it seemed to worsen. While curled up on my left side, it felt like I need to tap to make the inhale happen. (Which is actually very similar to how when I need to swallow, tapping helps. See Rice (2023).) Could this be a CCI/AAI/spine problem?

I was able to breathe fine while typing out my symptoms on my phone, but as soon as I tried to fall back asleep, my body freaked out like it didn’t know how to breathe.

I felt very unconfident about my breathing, in a way I hadn’t felt since maybe January or February 2024.

After about a total of an hour struggling in this way, I fell back asleep. A few times during the night afterwards, I woke up briefly to change my sleeping position, but my breathing was completely normal then. In the morning when I woke up, it was also completely normal. But as I sit up at my desk to type this up right now, I do feel a bit breathless, maybe from being shaken up by the experience.

It’s not clear if (1) my breathing was already a bit dysregulated, which made my Buteyko attempt fail, or (2) if I botched the Buteyko practice by just trying to do it on my own, and that failure dysregulated my instinctive breathing leading to worse breathing during the night. I would guess probably (2) is most likely. So from now on, I should only do Buteyko while I’m awake and guided by a recording. I’ve been having success with the Sasha Yakovleva’s “You Just Landed On This Planet” one and Neil Tranter videos (specifically Tranter (2024)), and hadn’t been experiencing any shortness of breath lately, so this was a scary surprise, and I learned how fragile my improvements were and how rigid my practice should be.

Another idea: my CP was critically low such that as I was inhaling I was already suffocating.

This experience seemed to wipe out all the progress I had made with Buteyko style exercises in the last several days. On the evening of 2024-06-08 I had a scary experience where I talked too much and my diaphragm/chest

became very tense, and I felt like I couldn't breathe. I started to go into a sort of shock, was shaking and felt very weak (couldn't stand). I thought I might die or have to go to the ER. But after maybe 30 minutes or an hour of just trying to take soft breaths through my nose, I calmed down. It's hard to say if my one misstep with Buteyko last night caused all of this stuff to happen (clearly, talking too much multiple times in the day also played a role), but I think it's important to explain what happened to me in detail to give some flavor of what can happen.

4 Different types of shortness of breath

The following is a typology of the *phenomenology* of shortness of breath (at least the ones i've experienced). In other words, I am distinguishing how shortness of breath can *feel* different to me, rather than trying to distinguish different *causes* of shortness of breath. Different causes can produce the same feeling (e.g. blocking your nostrils by mucus or by your fingers will produce the same shortage of air). And plausibly, the same cause can produce different feelings (maybe hypocapnia can feel like jittery anxiety on one day, but like brain fog on another day, depending on what else is going on with my body on a particular day). But here, I am just focused on the feelings, not the causes. Generally, the two will coincide, but the phenomenology is a coarser grouping, which I prefer since e.g. many different kinds of airway blockages can all feel the same and it makes sense to group all of those.

- CO₂ intolerance/overbreathing/hypocapnia: see The biological basis of shortness of breath
- psychological air hunger: where you are hyperfixated on your breath, and any even slight feeling of needing to take a breath (such as is natural right before you take a breath!) feels way too much air hunger.
- breath mistiming: this is the one where i'm breathing with a certain pattern of inhales and exhales, but i'm somehow doing it all wrong. probably i am consciously controlling the breath in ways that are bad (e.g. pre-empting the inhale). so even though i am "breathing", it just feels "bad" somehow.
- "central apnea": the one where i wake up gasping for breath, and where i feel like i stop breathing, and then my breathing rhythm is off where if i don't jiggle myself after the exhale it feels bad.

This (almost?) always happens in a hypnagogic state. (This is one of those rare cases where my unconscious breathing is off, so i have to consciously fix it, whereas the classical type of air hunger is where you consciously try to take control of your breath and this messes things up.)

- “suffocation”: where i feel like i’m not even breathing air, i’m breathing some other substance, so even when i take rescue breaths, it doesn’t feel good. you feel the air hunger even when you’re inhaling. 2025-01-23: nuts, I just had this type of air hunger, and described it to someone as “it feels like the air i’m breathing isn’t real air”; it was uncanny to see that i had described the exact same thing in the exact same way and then forgot about it.
- airway blockage: nose is stuffed, turbinates are enlarged, etc. See Clearing your nose.
- diaphragm/throat tension when talking/laughing/etc: see Talking out loud

2025-01-31: Example of one type of shortness of breath:

As I lie to sleep, anxiety builds at the end of the exhale. If I focus too much on my breathing, it’s like getting onto an escalator. I can’t do it. I need to tap with my foot or make some other movement for the anxiety to go away and to be able to inhale normally. When I messed up enough, I sat up gasping for air, even though I was breathing!

Air feels unreal. That feeling again.

For some reason, while writing this in a paper spiral notebook, my breathing is good. I think focusing on my breath make me subtly control it in bad ways (that I can’t even not do). So I must focus on other things.

Anyway, I haven’t had this in a while! So I am improving. It’s just that with this current AP, my symptoms are higher, I am more sensitive, and understandably this bleeds into my breathing patterns as well. But I know the way out. Keep up the soft nasal breaths. Use the tapping as a coping mechanism. If the air hunger builds too much, take a forced breath (preferably through the nose). Just keep going.

On the night of 2025-02-04, I did a 15 min Buteyko session (using my favorite Neil Tranter video) before brushing my teeth, and this actually helped me to *not* have the above while falling asleep (even though I have been having it every night since I wrote the above). Gentle Buteyko seems to just soothe the nervous system just enough that my breathing stops

being a problem.

4.1 Questionnaire

(I would eventually like to build this questionnaire up into a thing someone with air hunger can use to figure out which type of air hunger they have, and from there, figure out which practices would help them the most. But that's not what I have here, for now. For now, this is just some brainstorming.)

- Straining for air clears air hunger? Yes/No/Helps but doesn't eliminate air hunger
- Nose feels blocked or there is a feeling of resistance in the nose? Yes/No
- Happens as one is falling asleep? Yes, all the time/Yes, but not all the time/No
- Started when doing a breath hold? Yes/No
- Started when trying to exercise? Yes/No
- Started when trying to mentally exert? Yes/No
- Can silently breathe through nose at least a few times in a row? Yes/No
- Diaphragm tension? Yes/No
- Abs/core tension? Yes/No
- Throat tension? Yes/No
- Pulse ox O₂ sat reading
- Pulse ox perfusion index reading
- Where do you feel the air hunger? Chest/head/forehead/whole body/etc.
- Panic feeling at end of exhale? Yes/No
- Need to jiggle at end of exhale to avoid panic? Yes/No

5 The biological basis of shortness of breath

I *could* just tell you the exercises you need to do to fix your breathing, but personally I find it unsatisfying when people tell me to do a thing without also telling me why. So in this section, I will give a simplified but (as far as I know) accurate biological explanation of why you have shortness of

breath. This will help you understand *why* you should do the particular types of exercises I recommend later on (in Breathing exercises).

If you followed my advice and ruled out serious medical problems with a doctor, you may be thinking there is no biological cause of your shortness of breath, that it is more or less “in your head”. Certainly this is what doctors kept telling *me*, that it was all due to anxiety or some other hard-to-verify physical cause like mast cell activation (primary care physician, two pulmonologists, allergist).

It turns out that there is a pretty well-understood cause of air hunger! But to understand it, we need to have a decent understanding of the chemistry of respiration; in particular we’ll need to correct some misconceptions about breathing. So that’s what this section is about.

The aim of this section is to give you a useful mechanistic model of how respiration works. I will skip a lot of details because while it’s helpful to know them, they aren’t essential.

The central thing we’re trying to explain here is: why is it that if you have shortness of breath, the right thing to do is to breathe *less*, even though your body is telling you to breathe *more*? What is going on at the chemical level when people have shortness of breath?

(If you prefer a video instead of text, Cooper (2021) is the best explanation I have found. However there are a few things I don’t like about the explanation in that video, so I’ve written my own version here. But if you prefer a video explanation, then you can start with the video and then come back here. If you do rely on the video, please at least skim Rice (2024a) where I explain what I think Cooper gets wrong in his videos.)

5.1 How breathing works

(writing this section from memory; i think if i get anything wrong here, it will be a good thing to point out in the final version!)

How does breathing happen? here is a very simplified sketch:

The mechanical (inspiration and expiration): to inhale, your muscles cause the lungs to expand in volume. As the volume in the lungs increases, the air pressure inside the lungs starts to drop (relative to the baseline of atmospheric pressure), causing air in the atmosphere to enter your lungs, causing pressure to increase back toward atmospheric pressure. To exhale, your muscles *relax* which causes the lungs to contract (the lungs are elastic so they naturally contract); this decrease in volume causes an increase in

air pressure (relative to the atmosphere) so some of the air in your lungs leaves. It is also possible to do a forced exhale where the muscles actively contract the lungs in order to push out even more air, but this does not happen in a normal exhale.¹

The chemical: at the very edges of your lungs are tiny sacs called alveoli. the alveoli are surrounded by capillaries, and this is where gas exchange takes place. when air from the atmosphere arrives at the alveoli during inhalation, there is more oxygen in the atmosphere air, so the oxygen diffuses into the capillaries. At the same time, there is more carbon dioxide in the capillaries, so that diffuses back into the alveoli and then out into the atmosphere. this two-way diffusion is just happening all the time continuously, and inhalation and exhalation just bring or pump out the air so that more diffusion can take place. This two-way diffusion happening at the lungs is called **external respiration**. The oxygenated blood then moves through your body and goes out to all of your cells. At your cells, the blood has oxygen and your cells have carbon dioxide, so another exchange takes place, called **internal respiration**. Finally, all the while your cells are using up the oxygen to produce more carbon dioxide in a process called **cellular respiration**.

The neural: What causes the urge to breathe? Humans have what are

¹This footnote is intended to prevent a potential misconception. However, the actual details in this footnote are not important for understanding the rest of the document, so I've decided to not include it in the main text. Those with knowledge of chemistry will remember the ideal gas law, $PV = nRT$, which relates the pressure P , volume V , amount of substance n , and temperature T of a gas, with some constant R that varies depending on the units.

If we try to model the lungs naively based on this equation, we run into trouble. Let's see how that happens. During the exhale, the lungs contract by let's say 20%, i.e., its volume changes by a factor of $1 - 0.2 = 4/5$. To make up for this decrease in volume, now pressure must increase by a factor of $5/4$, i.e., the pressure increases by $5/4 - 1 = 25\%$. The increased pressure then expels the air, so n goes down, which then makes P go down again.

The problem is, when actually measured, the pressure in the lungs increases by only about 1 mmHg from a baseline of 760 mmHg (the atmospheric pressure), so the pressure in reality only increases by about $761/760 - 1 \approx 0.13\%$, not 25%. Why the huge discrepancy? It is because the exhalation does not happen in a rigid sequence like "the lungs contract all at once; the air pressure is raised; the air flows out to neutralize the pressure difference". Instead, as the lungs begin to contract, the air starts spilling out almost immediately, so as the pressure starts to build up from the contraction, pressure also starts to neutralize from the air escaping the lungs. So the actual pressure increase during exhalation cannot be calculated solely from the contraction in volume; it's instead coming from the *lag* from when the lungs begin to contract and when the air starts to leak out.

called chemoreceptors in their brain (and some other places) that tell the body to breathe. How do chemoreceptors know *when* to send the signal to breathe? Do they just have a timer in them and send a signal every few seconds? It turns out that they watch out for signals from the body, in the form of carbon dioxide levels, oxygen levels, and blood pH. Most of the time, when people breathe, it is because the carbon dioxide level increased above some threshold. Oxygen only triggers breathing when it gets really low (so low that I don't think most humans experience it except on very rare occasions). I think blood pH also matters, but maybe only during exercise? (but it might matter more for people with air hunger; see below).

Notice how the three aspects interact: the mechanical action allows the chemical exchange to take place (without just being stagnant at some dynamic equilibrium). The chemical process is what stimulates the nervous system. And the nervous system sends a signal to the mechanical part to make the motions.

5.2 Misconceptions about CO₂

From the above description, and based on your pre-existing knowledge, you might have the impression that CO₂ is just a waste product. but actually, both too-low and too-high CO₂ cause problems for the body. [TODO: say more]

i'm not very attached to "high levels of CO₂ are super healthy". i honestly haven't vetted the claims. but basically everyone (including mainstream western medicine) agrees that 35–45 mmHg is the healthy range for partial pressure of CO₂.

5.3 Chronic overbreathing and the bicarbonate buffer system

acid–base balance, bicarbonate buffer model (also called "renal compensation", e.g., see https://en.wikipedia.org/wiki/Renal_compensation and <https://youtu.be/G3YHBYVxmI?t=1527>)

mention how people with shortness of breath might not think they are hyperventilating. For example I don't think I've ever overtly hyperventilated in my life. And yet, I was constantly low-grade overbreathing. Chris Gilbert (2014, p. 68) says "Hyperventilation tends to persist in many people, and once a pattern of overbreathing is established it can be maintained by only a 10% increase in minute volume, which could be achieved by a

combination of deeper breaths, faster breathing, or an occasional sigh". Chaitow, Bradley, and Christopher Gilbert (2002, p. 246) is also great and I wish someone had pointed me to it because I kept thinking "there's no way I'm hyperventilating, I sure am not acting like how it is portrayed in media"; I have no idea why they removed it in the second edition of the book. However, I think their list of criteria for detecting overbreathing are not great. In particular, if you're nose breathing but your breathing is clearly audible, then I think that's a sign of overbreathing (this is what I was doing from like July 2023 to January 2024, and even beyond when I'm not being careful).

kidneys are supposed to adjust within 5 days. BUT mine seem to be taking a lot longer (it's been about 2 weeks of normal baseline CO₂ according to capnometer now as of 2024-01-31, and i'm still having air hunger daily). why could this be?

- it's possible that with my dysautonomia (and/or whatever else is wrong with me) my kidneys are just not NORMAL kidneys, but sluggish broken kidneys
- i also have an inkling that the studies they did on this was for ppl who overbreathed during the experiment, not ppl who have been overbreathing for literally over 10 years. it's possible (just like with SSRIs or various other medications or w/e) that if you've been overbreathing for much longer, then your kidneys are just "more stuck", "more set in their ways", than the standard experimentally verified number. if true, now that i am at a normal CO₂ baseline, i should just be able to literally keep going, keep doing the same thing, and the air hunger should just gradually disappear.
- maybe my bicarbonate levels take 3–5 days to adjust to one additional "rung", and there are like several rungs? i am not quite sure how that would be, but this could be why it's taking way longer than 5 days. (and experiments maybe only tested a "one-rung change".) It would make sense that making a greater change takes a longer time.
- the literature could just straight up be *wrong* about this, like, on what experimental or theoretical basis are they even claiming the kidneys take 5 days? i have not looked into the actual experiments so i can't tell whether people are just parroting flawed studies or the experiments were replicated.
- the other idea is that my CO₂ is totally normal now when I measure

during the day, but somehow, somewhere, I'm still overbreathing. Maybe in my sleep, maybe in small pockets of time when I'm stressed, maybe whenever I talk. And that little bit of overbreathing is preventing the bicarbonate levels from fully adjusting. Hence why I still feel subtle air hunger.

- Vipul notes that the literature may be basing kidney adjustments partly on mountain climbers ("altitude acclimatization") who are adjusting to low CO₂ so are shedding more bicarbonate rather than in the case of overbreathers who are trying to gain back their bicarbonate. Mountain climbers are also adjusting to lower O₂, not just CO₂. So it's possible that what happens to them is just quite different from what happens to chronic overbreathers.

I don't know why no one seems to ever talk about this, but there are literal blood and urine tests to measure bicarbonate levels (it's called "total carbon dioxide", and is included in the comprehensive metabolic panel, which is a pretty common test. Total CO₂ includes not just bicarbonate but also the dissolved CO₂ in blood and the CO₂ carried by hemoglobin or something, but the bicarbonate makes up most of the total carbon dioxide. However, I am still confused about how to interpret the results of this test). In the US at least, if you have a good insurance or Medicaid or something, then (assuming you can convince a doctor to order the test) this seems like a *way* cheaper way to see if you have hypocapnia, although it won't really allow you to do real-time biofeedback like with a capnometer. (Update(2024-02-07): this turned out to be WRONG. My total CO₂ has been normal this whole time despite my end-tidal CO₂ being low. So getting a normal level on the total CO₂ test does not actually rule out hypocapnia.) I think these blood/urine tests might also be very helpful in intermediate stages of recovery (like where I am at now as of 2024-02-02) where my CO₂ levels are normal according to the capnometer but I still have that subtle feeling of air hunger. (Update(2024-02-07): I am no longer so sure about this... I think the total carbon dioxide test might be useless.)

I have not tried this, but it is possible to *supplement* bicarbonate in addition to changing breathing habits, which might speed up progress: "Long-distance runners, sprinters and horse trainers have experimented successfully with doses of sodium bicarbonate, which supplements the natural bicarbonate buffer and opposes the lactic acid load created by exercising muscles" (Chris Gilbert 2014, p. 66) <https://www.ncbi.nlm.nih>

.gov/pmc/articles/PMC435634/pdf/jcinvest00601-0100.pdf (“STUDIES ON HYPERVENTILATION. II. ELECTROCARDIOGRAPHIC CHANGES IN NORMAL MAN DURING VOLUNTARY HYPERVENTILATION” by B. CHR. CHRISTENSEN) also mentions an experiment to give large doses of sodium bicarbonate to subjects to generate (!) a feeling of air hunger. more bicarbonate mimics the effects of less CO₂. so maybe supplementing bicarbonate isn’t the solution.

2024-05-07: so something interesting has happened: on Sunday (2024-05-05), all day long, my parents and i did a lot of cleaning and moving furniture around the house, which created a *lot* of dust, and because i didn’t seem to be reacting at the time (e.g. I wasn’t sneezing, or maybe I only sneezed once) I didn’t think to wear a mask. but yesterday i had some more classic allergy symptoms, like sneezing, breathing difficulty, and itchy-feeling insides. and today i am having breathing difficulty again. but what’s interesting is that this breathing difficulty feels *exactly* like the breathing difficulty i had in 2023, before the capnotrainer stuff. and specifically, i’m noticing that using the nasal strip is helping a lot but the saline spray did nothing. but i don’t think i was mysteriously exposed to a lot of allergens in 2023 that somehow went away in 2024. so instead i am thinking that allergies have caused my turbinates to become swollen, and completely via a different mechanism, overbreathing also causes my turbinates to become swollen. which is why the nasal strips worked so well even in 2023. so, what’s the point of all of this? i think one takeaway i have is that it’s *not* just the Henderson–Hasselbalch stuff and increased CO₂ sensitivity in overbreathers that causes the shortness of breath feeling, but also, there’s a pathway like overbreathing → body tries to prevent you from breathing so much → inflates turbinates to compensate → now your nostrils are super tight → feel like you can’t breathe.

maybe the turbinates get puffed up during allergies to prevent more allergens from coming in?

also it’s interesting that the breathing difficulty came on only 1–2 days *after* probably most of the dust exposure. this actually also happened in 2023, i had a breathing attack in the early morning of 2023-02-15 while i was asleep, and had to go to urgent care later that day. but the actual exposure (and at the time i didn’t know it was from this, although it was one of the ideas i had) was on 2023-02-13 when in the evening i cleaned my window blinds (which hadn’t been cleaned in a very long time and had a lot of dust). i remember mentioning the blinds dusting to one of

my pulmonologists, who dismissed the idea given that it was two days after the exposure. but now that it's happened *again*, after my breathing has substantially improved (and with no other reason for my breathing to suddenly get much worse, e.g. i wasn't talking a lot), i am sure it's from the dust/allergic reaction (although i'm less certain about whether it's some sort of systemic reaction, or if it's mediated by the turbinates inflating).

i wonder if in cases like this where allergies are clearly involved whether azelastine will actually have any benefit. however, i seem to not respond to any of the oral anti-histamines that i have tried (claritin, xyzal, zyrtec) even when I am in this state.

as of 2024-05-08 my breathing is basically back to my post-capnotrainer usual baseline. i haven't needed to use a nasal strip today.

5.4 The Henderson–Hasselbalch equation and CO₂ tolerance

[argh, this is kind of hard to explain well. curse of knowledge...]

Okay, so CO₂ levels (partial pressure?) in the blood regulate breathing. Now let's imagine what happens when someone overbreathes. You might think, okay, CO₂ goes down, so then you don't get as much of an urge to breathe. That's true in the short term. But if someone *chronically* overbreathes, then the body tries to correct this. You see, the body tries very hard to keep the blood pH at a narrow range of 7.35–7.45. If one chronically overbreathes, then since CO₂ causes the blood to become acidic, the *lack* of CO₂ (you're overbreathing, remember) causes your blood to become too alkaline. So the body compensates by trying to make blood more acidic. How? The kidneys dump out more bicarbonate (HCO₃⁻) into the urine. Since bicarbonate is alkaline, getting rid of it makes the blood more acidic. The kidneys don't react immediately; all this happens over the course of hours to roughly 5 days.

[... some more stuff here ...]

The important point is that because of the form of the Henderson–Hasselbalch equation, the more CO₂ you have, the less sensitive pH is to slight increases in CO₂. So breath holding for a few seconds does not cause air hunger/panic/suffocation. Whereas if your CO₂ is already too low (hypocapnia), then your pH is still normal (thanks to the bicarbonate levels being adjusted by your kidneys), but because of the low CO₂, even slight changes in CO₂ can cause large shifts in pH. I think it's this decrease in pH (when CO₂ rises, blood becomes more acidic so pH goes down) that

triggers a strong sense of air hunger. If you give in to this urge and breathe, then your pH will be restored, so *temporarily* you will feel better. But if you keep giving into this urge and breathing more and more, then over time you're shifting down your bicarbonate buffer even more. Your body adjusts. But that means your baseline CO₂ also keeps decreasing. So now you're *even more* sensitive to slight increases in CO₂. It's a vicious cycle. (and this isn't even talking about the negative health effects of low CO₂.)

Let's look at some numbers.

$$7.4 \approx 6.1 + \log_{10} \left(\frac{24}{0.0307 \cdot 40} \right)$$

above are the values for a normal person.

When I first used a capnometer, my CO₂ was around 25 mmHg. So to get the same pH, my bicarbonate level must have been around 15 (here I'm just solving for x in the equation $24/40 = x/25$ in order to make the input to $\log_{10}$ the same):

$$7.4 \approx 6.1 + \log_{10} \left(\frac{15}{0.0307 \cdot 25} \right)$$

Now let's say both individuals do a breath hold to raise their pCO₂ by 5 mmHg. In the case of the healthy person, their pH goes down to 7.34, but in the chronic overbreather's case, it goes down to 7.31.

Here's the other thing Vipul showed me:

Suppose your equation right now is:

$$\begin{aligned} \text{pH} &\approx 6.1 + \log_{10} \left(\frac{[\text{HCO}_3^-]}{0.0307 \cdot \text{pCO}_2} \right) \\ &= 6.1 + \log_{10}([\text{HCO}_3^-]) - \log_{10}(0.0307) - \log_{10}(\text{pCO}_2) \end{aligned}$$

Now you hold your breath so that the pCO₂ goes up by some amount, ΔpCO_2 , and the pH changes by some amount, ΔpH (it's a negative quantity, since your pH goes down when the blood becomes more acidic). Your equation now looks like:

$$\begin{aligned} \text{pH} + \Delta\text{pH} &\approx 6.1 + \log_{10} \left(\frac{[\text{HCO}_3^-]}{0.0307 \cdot (\text{pCO}_2 + \Delta\text{pCO}_2)} \right) \\ &= 6.1 + \log_{10}([\text{HCO}_3^-]) - \log_{10}(0.0307) - \log_{10}(\text{pCO}_2 + \Delta\text{pCO}_2) \end{aligned}$$

Now let's subtract the first equation from the second; we get:

$$\begin{aligned}\Delta\text{pH} &= \log_{10}(\text{pCO}_2) - \log_{10}(\text{pCO}_2 + \Delta\text{pCO}_2) \\ &= \log_{10}\left(\frac{\text{pCO}_2}{\text{pCO}_2 + \Delta\text{pCO}_2}\right) \\ &= -\log_{10}\left(1 + \frac{\Delta\text{pCO}_2}{\text{pCO}_2}\right)\end{aligned}$$

What does all of this show? Well, the quantity $\frac{\Delta\text{pCO}_2}{\text{pCO}_2}$ is actually the fractional change in pCO_2 . So it shows that *the change in pH is a function of the fractional change in pCO_2* . This is one way to understand why having low CO_2 means your pH changes more easily: the same amount of absolute CO_2 change results in a greater fractional change in CO_2 . (Note: fractional change is almost the same thing as percentage change, just divided by 100.)

Here's a model you can play around with in Python:

```
from math import log10

# These two values are obtained from measurements
etco2 = 35 # This comes from the capnometer
total_co2 = 28 # This comes from the "total carbon dioxide"
               # blood test

# Arterial pCO2 is typically a little bit higher than the
# end-tidal pCO2. Here we just guess it to be 5 mmHg higher.
paco2 = etco2 + 5

carbonic_acid = 0.0307 * paco2

# total CO2 = bicarbonate +
#             CO2 as carbonic acid +
#             CO2 in hemoglobin (estimated to be 10% of the
#                               total CO2)
# So we are just solving for the bicarbonate term:
bicarb = 0.9 * total_co2 - carbonic_acid

# This is the Henderson-Hasselbalch equation for pH
pH = 6.1 + log10(bicarb / carbonic_acid)
```

print(pH)

This seems wrong to me, but I should look at the cited works by Ley: “Ley (1988a, 1988b) has offered a plausible explanation for this phenomenon based on the body’s adaptation to chronic hyperventilation. The long-term reduction in bicarbonate buffer concentration, offsetting lowered PaCO_2 , returns pH to normal, but the equilibrium is an uneasy one, dependent on the hyperventilation continuing. This situation makes the individual more susceptible than ever to a rise in CO_2 (more acidity) because the alkaline buffer has been reduced. Consequently a change in breathing toward normal, away from hyperventilation, would feel closer to suffocation than if the person possessed normal bicarbonate buffering capacity.” (Chris Gilbert 2014, pp. 68–69) Specifically, I don’t think it’s the lower bicarbonate that is the problem, it’s the lowered CO_2 . I think the literature is very misleading about this point and they all keep repeating the same mistaken thing, and none of them actually provide a mathematical model of this. In more detail: The specific sentence I am critiquing is: “This situation makes the individual more susceptible than ever to a rise in CO_2 (more acidity) because the alkaline buffer has been reduced.” In the HH equation, the bicarb concentration is a constant because the rise in CO_2 being mentioned here is an acute one. The things that are changing are the CO_2 and pH, and the reason the pH changes more is because the CO_2 is already low. The reason the CO_2 is low is because the person chronically overbreathed. The reason the CO_2 *remains* low is because the bicarbonate level also changed. But you see how that’s like, several levels of indirection until we even get to the bicarb. Saying the word “because” followed by the bicarb/alkaline buffer makes it seem like that’s the direct/primary reason, but in reality it’s instead just this thing in the background that is keeping up the equilibrium. However, I do admit the author here may have had the correct model and was just wording it poorly.

“Thus the chronic hyperventilator’s pH regulation is finely balanced: diminished acid (the consequence of hyperventilation) is balanced against the low level of blood bicarbonate maintained by renal excretion. In this equilibrium small amounts of over-breathing induced by emotion can cause large falls of carbon dioxide (PCO_2) and, consequently, more severe symptoms. In other words, the chronic hyperventilator lives closer to the threshold for hypocapnia than the normal person.” <https://web.archive>.

org/web/20230127113032/https://buteykoclinic.com/docs/hyperventilation%2Ba%2Btherapists%2Bpoint%2Bof%2Bview_1401456256.pdf – another example of a paper seemingly getting this wrong.

for advanced buteyko practitioners however, my current guess is that their PCO₂ levels are normal (40–45 mmHg). so now, how do we explain how they can hold their breaths for so long? i think some other factors are at play here including:

- less sensitive chemoreceptors
- hemoglobin count or efficiency
- myoglobin count or efficiency
- reduced or more efficient cellular metabolism (slower rate of generation of CO₂)
- higher lung capacity

some of the above may also be happening in going from chronic overbreather to normal breather, but it's not obviously necessary to explain the differences in CO₂ tolerance between overbreather and normal.

5.5 Upshot

So what we've established is:

- If you have shortness of breath with no clear medical cause, it is likely that your CO₂ level (and thus bicarbonate level) is too low.
- Low CO₂ means you are more sensitive to changes in CO₂, in particular to increases in it. This is why you always feel short of breath.
- The solution is *not* to “take a deep breath” and get more oxygen (you already have plenty of oxygen, probably too much, as can be verified easily with a pulse oximeter), but rather, to breathe *less* so that you *gradually* raise your CO₂ levels. The emphasis is on the word “gradually”. If you just hold your breath as long as possible in the hopes of raising your CO₂, you will just give yourself a panic attack. The kidneys take time to adjust the bicarbonate levels. So you want to sit with a slight-to-moderate amount of air hunger, over the course of days and weeks, so as to gradually raise your baseline CO₂. How to do this? We'll discuss breathing techniques and breathing habits more elsewhere in this document (see the sections Techniques vs habits and Breathing exercises).

6 What does normal breathing feel like? Is healthy breathing the same for everyone?/Is normal breathing universal?

One of the weird things I discovered while using the CapnoTrainer is that the breathing style that brought CO₂ levels up to normal in the most effortless way turned out to be what the Buteyko people have been saying all along: silent nasal breathing. This raises the question of whether there is some universally “optimal” breathing style that everyone should have, or if it’s more of a preference (different people can have different breathing styles, but what matters is that your CO₂ levels are normal and you don’t have air hunger or other breathing issues).

As I said in the previous paragraph, the Buteyko people believe there *is* such a thing as universal optimal breathing, namely effortless silent nasal breathing that is invisible from the outside. However, Buteyko people don’t prescribe a certain rate of breathing (e.g. they don’t say you need to breathe in to a count of 4 seconds and exhale to a count of 6 seconds).

Peter Litchfield on the other hand says that it doesn’t matter how you breathe – you could even mouth breathe – as long as your breathing habits are “functional” (by which I think he means your CO₂ levels are at like 40 mmHg). For example see Litchfield and Reamer (2023, p. 21), where “There is an ideal breathing pattern that each of us needs to learn” is listed as a misconception.

Unfortunately it’s hard to get descriptions of what normal breathing feels like from normal people because those are the people for whom nothing about their breathing has ever been wrong. so they don’t think about their breathing. they don’t know how to contrast normal breathing to what abnormal or overbreathing feel like. (It’s sort of like how native speakers of a language are often the worst teachers of the language because to them the quirks of the language are natural and they’ve never known what it’s like to *not* know the language. To learn a language, you need to find the non-native speakers who have managed to achieve proficiency in the language because they are the ones who know all of the struggles you are going through.) here i will try my best to share glimpses of what normal breathing feel like, although caveat as usual that i have not fully fixed my breathing problem yet.

In normal breathing:

- There is no urge to take a deep breath as long as one is just quietly sitting or doing some “minimal” activity such as standing, walking a few paces, or talking at a normal volume. This is indefinite, in contrast to air hunger where one might be able to take a few quiet breaths in a row but then the urge to take a deep one builds up over time. in normal breathing, there is no buildup.
- silent nasal breathing is sufficient (at least when seated or lying down)
- it is still possible to take a deep breath. these feel better than normal breaths, but there is no compulsion to keep achieving that deep breath. it’s currently not clear to me why normal people don’t develop an addiction to taking deep breaths... I think it may be similar to drinking water or something: imagine a world where people normally get water drop by drop; then it might feel particularly good to drink a big gulp of water; but if you keep gulping water, you just get too much of it and you reach a point of satiation.

7 Can you overshoot your CO₂ levels by doing too much reduced breathing?

so you combat air hunger by breathing less, which raises CO₂. what if you just keep going? could you be overshooting and getting *hypercapnia* instead?

Here’s a heuristic argument for why this shouldn’t happen: even people who overbreathe don’t overbreathe so much that their CO₂ just keeps dropping indefinitely. Each style of breathing seems to asymptote at a certain point. So I think Buteyko-style reduced breathing also asymptotes in terms of CO₂ levels.

But also, I think there must be some other effect going on, because I don’t think the Buteyko people have abnormally high CO₂ (that’s just my current guess). the only source i’ve found so far is the graphs in Altukhov (2009) where the highest tier (tier V) under “Super Endurance” has an alveolar CO₂ of 54 mmHg. But they also list “normal” as being 46 mmHg which seems too high...

the main lesson here is that having a capnometer is quite useful, to give the assurance that your CO₂ levels are normal.

another reason to not worry: if you are doing buteyko *correctly*, then any reduction in breathing must come from relaxation. you should only just

barely have any air hunger. if you are overshooting and making yourself actually uncomfortable, then you will start to feel a lot of air hunger and tense up, and that will be a signal to ease off. so basically your body has a corrective mechanism built in.

8 Is Buteyko breathing a cure or does it just mask the root cause?

I think Buteyko style breath holds may be harmful in that it's compensating for your poor breathing by increasing your CO₂ tolerance. But that's *not* what's actually wrong with your body (unless you just have low CO₂, in which case Buteyko breathing can be a cure; in my case I think there may be something neurological, or something with the vagus nerve, or something with oxygen not getting delivered to cells, or something like that, while my ET CO₂ is now normal after doing capnometry biofeedback). So you can do Buteyko if you're already healthy, and get some benefits, but doing it when you're ill may mask your symptoms without treating the root cause.

9 A short timeline of my improvements

i don't know how representative the speed of my improvements is, but it's the only case i am deeply familiar with, and it might give people an idea of what to expect.

- sometime in 2010–2015: i go to my primary care doctor several times complaining about shortness of breath. each time he gives me an albuterol inhaler and i try it and it does nothing. then the problem just randomly goes away and i give up trying to figure out what's going on.
- 2019: the breathing problem is more persistent starting around this year, which is also the year when most of my other symptoms started showing up.
- in early 2022, ML gives me a brief overview of CO₂ theory but i don't understand it at all (to be fair, it was an impromptu explanation and also i was probably brain fogged). but he gives the upshot that the concrete thing i should do is to briskly walk for 40 minutes per day for one week straight, and this should normalize my CO₂ tolerance. i

try it, and it does nothing. he also mentions buteyko breathing to me, in probably in mid-2022, saying i can maybe look into it. i don't (not because i didn't want to, but because i have so many other symptoms to research and deal with and quite limited energy due to illness and because i read a little bit about it and it seemed way too difficult to understand).

- 2022–2023: a lot of going to doctors to try to figure out what's wrong with my breathing. multiple chest x-rays, methacholine challenge, pulmonary function exam, exhaled nitric oxide test, a bunch of doctors (including pulmonologists) listening to my breathing with a stethoscope.
- July 2023: emergency room visit due to severe difficulty breathing. this happened after like a 2–3 hour doctor appointment where i had to talk a lot. i had to cut the meeting short because i was unable to talk by the end. going to the ER really motivates me to solve the breathing problem, because i don't want this to happen ever again. i start doing a lot of digging on the internet about air hunger etc.
- July/August 2023: i learn a basic version of the CO₂ theory via howard cooper videos. i realize this is probably the thing ML was trying to tell me about in early 2022. i start playing around with breath holds and other gentle buteyko exercises using neil tranter videos. in end of August i also started reading Nestor (2020) and trying to switch to nasal-only breathing, and basically succeed over the course of a month or so. i also probably tried nasal strips for the first time this month.
- (not much new happens in this period in terms of my understanding, but once i make the adjustment to nasal breathing around August–September, my breathing problem is generally like 60–80% better, except when i speak, at which point it goes back to being moderately bad.)
- January 2024: capnotrainer. i finally figure out how to silently breathe. my baseline end-tidal CO₂ goes from like 27–30 mmHg to 37–40 mmHg. there's just a subtle air hunger left now, except when i talk; when it talk it can still get pretty bad, but i'm better at recovering by doing silent relaxed nasal breathing.
 - 9th: first day of capnotrainer.
 - 15th: breakthrough.

- February 2024: simon spire course; no progress.
- March 2024: starting to do a more serious buteyko practice, as i think my main problem right now might just be that i'm not logging enough time doing deliberate breathing exercises.
- May 2024: room swap creates a lot of dust, my breathing is worse for like a week afterwards. This is when I realize for sure that dust dose make my breathing worse if it's in large quantities. However, the mechanism seems to be via making my turbinates swell up, rather than having a more systemic reaction.
- June(?) 2024: the diaphragm tension stuff gets worse and worse, to the point where I have to go mostly mute because talking even a little bit can really aggravate it. On June 8th I have a major episode where I feel like I need to go to the ER again, but I somehow scrape by by doing soft nasal breathing to recover.
- August 2024: a few days of perfect breathing (but I still can't talk very much).

10 Sequence of cumulative constraints of breathing practice

We are now transitioning to the more practical sections of this document. But before we get to the actual exercises, I want to say a few things about the mindset/global structure of the practice.

I think it may be helpful to conceptualize breathing practice² not as a set of "breathing techniques" that you practice over time to help calm you down or whatever (such as: "when you're feeling anxious, do square breathing; once you feel comfortable with that, you can move onto different patterns like 4-7-8"), but instead as a *sequence of cumulative constraints*. The reason for having a *sequence* is that it's too hard to pay attention to the more subtle stuff and get it right all on the first try before you get the basics down; each stage might take a month or more. (The other reason for this particular sequence: if you follow the sequence, then by doing the previous steps, you automatically almost do the next stage, so that at each stage you only need to put in a *little* bit of effort, not invent a whole new thing.) The

²By "breathing practice", here I specifically mean breathing practice to cure shortness of breath from over-breathing, but it may also apply to meditative practices as well, I don't know.

reason for having *constraints* rather than practices is that it mostly doesn't matter *how* you achieve good breathing, you just want the outcome.

This page used to use the phrase “sequence of moving targets” rather than “sequence of cumulative constraints”, but I renamed this page because I thought the new version is a more accurate phrasing. Like the Gram–Schmidt process in linear algebra the constraints are cumulative (at each step n , you want to make sure that the new vector you create is orthogonal not just to the $(n - 1)$ th vector, but to *all* the vectors $1, \dots, n - 1$. This kind of mechanic also happens a lot in puzzle games where if you get one thing in a “good” state, and then try to get another thing in a “good” state, the first one will turn back into a “bad” state, and the difficulty of the puzzle is to make sure you can get all the good states cumulatively). Also “constraint” feels more accurate than “target” because you’re not trying to hit some new breathing pattern some of the time, but rather you’re trying to *only* do the “good” kind of breathing *all* of the time.

10.1 The sequence

Here is the sequence I am thinking of:

1. Initially, because I was still mouth-breathing a significant percentage of the time (20%?), my constraint was to do nasal breathing. *Any* nasal breathing. The goal was simply to get away from the habit of mouth breathing. Nasal breathing is a good first target because (a) it's extremely obvious whether any particular breath happens through the nose or mouth; and (b) on its own it already has pretty dramatic effects (see Nestor (2020) for a lot more detail on this). Achieving nose breathing itself occurs in two stages:
 - Consciously do nasal breathing. You use any and all tools you have to try to nose breathe: just anxiously paying attention all the time, nasal strips (nasal splints), nose cones (nasal cones), nasal dilators, azelastine, NeilMed, allergy sheets, etc. (Flonase is another option that *might* help people but it didn't work for me because of side-effects and there's a chance it builds up a dependence even though it's not supposed to?) You want to get to a point where the only times you mouth breathe are when you have to blow your nose, when you swim, when you take a shower, or things like that. I nose breathe even with a mask on,

even when I'm brushing my teeth, and as much as possible even in my sleep (mouth taping or a chin strap might help with this). (This step took around 1 month for me.)

- Eventually, it becomes automatic, and you won't even need the training wheels. (After a month, my nasal breathing became habitual/automatic. I remained at this stage for the next 4 months or so, not because I wanted to but because I wasn't sure what to do next.)

2. Next stage: silence your breath.

- Same two stages here (conscious/manual techniques, then unconscious/automatic habit). Silent breath is the next constraint because it's also easy to detect: it means you can't hear your own breath at all (neither inhale nor exhale). Silent breath. At least when you're just sitting or standing. I'm not sure how feasible it is yet to have a silent breath while exercising. (This stage took about a week once I got the CapnoTrainer, though arguably it's still ongoing. In particular, there was a 24 hour period where I had a kind of epiphany and figured out how to do this right.)
- As with the nose breathing stage, you don't have to get it to be perfect. Sometimes, breathing in a silent way can make your diaphragm or your throat hurt. I think this happens because you're manually trying to control those muscles too hard by tensing them up in the wrong ways. If that's the case, you can try for a while to just make a bit of noise when breathing, just like you might still have "rescue breaths" through your mouth sometimes.
- Eventually, given a choice between silent vs natural, you want to pick natural, but still silent is the initial target because it's so much easier to tell when something is silent vs natural.

3. Next stage: smooth your breath.

- Even when your breath is through the nose and silent, it can be "jagged" on the CapnoTrainer. It can even *feel* jagged to you, like you're manually forcing air out in little bursts, instead of it just flowing out like it should. I don't know how to fix this one yet, but it seems harder to notice and harder to fix, so that's why it comes later. But the idea, at least for the exhale, is to just try to

let go of the breath and naturally let your diaphragm relax. It should feel like your body is just relaxing, instead of exhaling (even though of course you are also exhaling). As I said, I'm still experimenting with this one.

- This one is tricky because I notice that as I try to do it my diaphragm can get sore if I do it wrong. In particular, it can sometimes help to break the "silent" rule in order to get the breath smoother. There's a distinction between breathing in a way so that you aren't hypocapnic, and breathing in a way that is comfortable and not painful to your body. Every combination is possible.

4. Next stage: release tension in diaphragm/throat? this might be a specific-to-me thing, but even after my breath seems smooth enough, my diaphragm and throat get really tense. i think i must be artificially tensing these parts up to make my breath more quiet or something, but i don't consciously feel like i am exerting any effort. so it's confusing.

Note that the above is essentially (some of the particular techniques taught in) Buteyko breathing. The differences are: (1) pedagogical, in that I think focusing on each step in sequence works a lot better, and (2) whether to use a CapnoTrainer to help with some of the steps.

10.2 Any more stages?

i have a feeling that the final stage (beyond the ones i listed in the previous section) is "exercise while doing all of the previous ones". this is explicitly part of some buteyko guides. vipul also (i think independently) recommended i do this.

upper chest (intercostals) vs diaphragm? i feel like this one happens automatically by doing the ones above. But for some reason this is like the focus of many breathing gurus/meditation experts. but it's not necessary! in fact, i think it can be harmful to focus on your diaphragm too much – it leads to exaggerated/forceful movement of it which can cause overbreathing/sore muscles. like, sure, it's fine to check your abdomen to see that you are generally breathing with your diaphragm more than upper chest, but it shouldn't be a target.

deprecated: slow down your breath.

- Your breath should already be slower if it's more quiet and smooth. But when you're trying to make it quiet or smooth, you don't want to focus on slowing it down ... that just makes you uncomfortable because you're not used to that much air hunger. But once your breath is silent and smooth, it's time to focus on slowing it down. I still haven't gotten to this stage so I don't know how slow is optimal. I suspect this is the point where an HRV monitor or such would be handy: there might be individual variation in the optimal rate. Notice how it's much harder to say what "slow breath" is compared to "nasal breath" or "silent breath" or "smooth breath", and it is much harder to track without an instrument like a capnometer (counting in your head all the time seems annoying!).
- UPDATE (2024-01-26): i am thinking more and more that slowing your breath is unnecessary; it's like the diaphragm thing and it automatically happens. i think some of the buteyko people also say the same thing. e.g. see Andrey Novozhilov (2012); in the section "Approach 3 – Just hint at reduction of airflow" there is an illustration of a snake where slowing down the breath is a sign of doing the exercise wrong. Under mistakes, he lists "Increasing the frequency of your breathing or changing the length of inhalation or exhalation." (2024-06-17: I now think that temporarily playing around with your frequency of breathing is actually sometimes very helpful. I had such an experience on the night of 2024-06-16. But I still agree that long-term, or even just outside of a scheduled session, you shouldn't have to adjust your rate of breathing consciously; it will take care of itself. See also: note from 2024-07-18.)

11 Clearing your nose

In my view, nasal breathing is essential for overcoming shortness of breath (once one gets used to nasal breathing, one realizes that any breath taken through the mouth feels unnatural to some extent). It's also the very first step. There's a lot of strategies here so I'll list a bunch along with what I think of them. My guess is that different ones will work for different people, so you kind of want to try a bunch and see what works (most of these are quite cheap).

It's very important to keep in mind that most of these strategies (namely

nasal wash, mechanical dilation, and medication, but not Buteyko-style breathing and nose unblocking exercises) are *slightly evil short-term hacks*. They aren't *hugely* harmful when used occasionally or in the early stages of fixing your breathing pattern, as far as I can tell, but I think they do cause some harm in the sense that closed up nasal passages are a signal from your body that you are overbreathing, and by forcing your nose open, you are disregarding what your body is trying to tell you. It would be as if you had a cold and you were feeling fatigued (i.e. a signal from your body to take rest), but you drank a bunch of coffee and took stimulants so that you could work out anyway. So you don't want to just keep relying on these strategies long-term; your goal should be to improve your normal, unconscious breathing to the point where you're always doing silent nasal breathing. But early on, as a way to avoid mouth breathing (which is basically like mechanical dilation on steroids, so even worse than the hacks in this section), I think these strategies are fine to use.

I won't say much about *why* it's better to breathe through your nose, as I think this is already covered pretty well in other sources, in particular Nestor (2020), which gives a pretty vivid description of the author's own experiment of breathing solely through his mouth (it's scattered throughout the book, so unfortunately I can't cite a specific chapter).

I think there are two broad ways you can react to "you need to clear your nose". One of them will be to say "Yeah, I hardly ever breathe through my nose; I know I need to work on it but I don't know how". The other is to say "I already breathe through my nose; I don't think I need any advice". For this second type of person I want to mention my own anecdote. Before I started really focusing on my breath, I used to think I was already a nasal breather, but when I started paying attention I realized that I was actually taking quite a few mouth breaths throughout the day. A sure way to find out is to tape your mouth during the day (even for just 15 minutes or so) and see how easy it is for you. I think I probably went from around 80% nose breaths to 99.9% nose breaths by following the strategies listed below (just a wild guess unfortunately because I didn't keep good records); during breathing attacks, I probably went from 10–50% nose breaths to 95% nose breaths. So if you already think you are a nasal breather, I would still try paying attention.

The more I work on my breathing, the more I realize that having a good nose (opened up, not full of mucus) is key to good breathing. See also the section on Allergies for more thoughts.

11.1 Strategies

Here are the main strategies I recommend for clearing your nose so you can nose breathe more:

11.1.1 Nasal wash

- Saline spray: works less well than the saline wash, but it's a lot less unpleasant. it doesn't go "deep" enough, which is why it is both less painful and doesn't work as well. i went through about 1.5 bottles of the NeilMed one (lime green color bottle) before my nose improved enough that I didn't need to use it anymore. 2024-07-12: more recently, for whatever reason i've had times where i've felt more "resistance" in my nostrils, and at those times i've found the spray useful again. i only need it about once a week or once every 2 weeks. i have found that one time when i got a brand new bottle and tried to really get the water in deep, it actually backfired and made my breathing worse. what i like about the spray is that it's less painful but it's also more easy to titrate how much force you want. today i just did some light wetting of my nostrils and then lightly blew my nose, and i can tell my breathing is more relaxed (when your nostrils get narrower, you have to use more of the muscles in your throat, upper chest, diaphragm, etc, to get enough air). Sometimes you can feel dry mucus in your nose, or "stuck" mucus in the back of your nose (where you can't even blow it out with just a tissue paper, because it's "stuck" or "too thick" somehow), and in these cases I've had good luck using the spray to loosen up the mucus enough that it can then be blown out. On *some* bottles, the spray can be very strong at first (there seems to be some manufacturing differences between bottles even if you get the same kind, so some bottles seem to be much more pressurized than others, in my experience), but as it loses pressure inside the bottle, the strength gradually weakens. So if it feels like the inner lining of your nose hurts when you use it at first, just do a little bit at first (or try to do it from slightly farther away, although this will spray salt water over your face...) until it loses enough pressure that the strength is just right.
- Saline wash/netipot: tried this once. it works really well for clearing your nose, but it was also quite unpleasant. make sure to never use

tap water for this.

11.1.2 Mechanical dilation

- Nose strips (also sometimes called nasal strips or nasal splints): these were quite helpful. at first i used the RiteAid brand ones and they worked well enough, and then i got cheaper ones from Amazon and those also worked fine. so my guess is that there isn't much of a quality difference between different brands, and you can just go with whatever. the main thing to watch for is that your skin should not get irritated from using them. also, you should *immediately* notice an improvement. if you don't immediately breathe better, then you're either not putting them on right or the thing isn't working. There is also a magnetic strip made by a company called Intake Breathing. I never used it so I don't know how well it works, but if you can't get regular nose strips to work, you can look into this.
- Nose cones (also called nasal cones): these didn't work for me (i got them because i was encouraged by how useful the nose strips were, and cones seemed like reusable strips). they don't go deep enough in the nose, so they really only help i think if you have a deviated septum or something. . .
- Nasal dilators: I didn't try these, but you can check out this video for some examples. Note that the video doesn't use the term "nasal cones" and instead just calls everything "dilators", whereas in this document I've distinguished cones from the more "clip-like" dilators. The video mentions that these cones/dilators can become covered in mucus – but that's a signal from your body that you're overbreathing, and the mucus is trying to clog up your nostrils to make you breathe less! So watch out for signals like this from your body.

11.1.3 Buteyko-style breathing and nose unblocking exercises

- Patrick McKeown's nose unblocking exercise: this is one where you pinch your nose while holding your breath, and swaying or walking around as you do that. I tried this back when my nose was stuffier, and it didn't work very well for me, but i also didn't try very hard to get it working (i just followed one of his videos). 2024-07-13: tried it again and i don't think it worked. when my nose is doing worse, i

am already having breathing problems, so holding my breath is hard, and i end up gasping for air after the breath hold is done, which i think just decreases my CO₂ further, defeating the point of the breath hold. maybe i need to do a bunch of small 1–2 second breath holds.

- Just breathing less and not taking mouth breaths. See The better your breathing, the better your nose. This don't work in the very short term (of the next few hours), but it can work surprisingly quickly. I was able to fix one of my "nose keeps getting plugged up in the mornings for the last 2 months" type issues in the course of 2–4 days. (However, this was already after I had acclimated to silent nose breathing *most* of the time.)

11.1.4 Antihistamines and steroids and allergies

- Antihistamine spray (e.g., azelastine): my guess is azelastine did nothing for me. but maybe it helped a bit with reducing mucus production during the night. i never tried olopatadine (the more famous antihistamine nasal spray) because my insurance doesn't cover it.
- Antihistamines in general (e.g. Zyrtec, Claritin, Xyzal, Allegra, etc.): see the section on Allergies for more. You should probably try all the ones (one at a time) just to see which ones work best for you. Also I think for me the effects of antihistamines build up over the course of a week or so, so you should probably try using it for a week straight at full dose and see how your breathing (especially nasal resistance) changes over the course of the week.
- Steroid spray (e.g. Flonase): i would stay away from these, unless you already know they work well for you. definitely try all the other ones in the list first. note that there are different kinds of steroid sprays and some of them are *very* bad to keep using long-term. Flonase is supposed to be one of the safer kinds, but i only used it once so i can't speak to any longer-term effects.
- Washing bedding (especially(?) if you have dust mites allergy). Especially if the nose restriction feeling is worst in the morning and then gets better over the course of the day.

11.1.5 Other tricks

- sometimes just sniffing out with my nose in a slightly more forceful than usual way makes something shift inside my nose and I feel less air resistance.
- Similarly to the previous point, lying down or sitting up can make mucus flow (I had this happen on 2024-07-22, where my nose was feeling dry and it was hurting, but then lying down made mucus flow and I could swallow mucus and then my nose felt fine). Sometimes all you need to do is change posture and your nose will be happy. On 2024-08-06 I had clogged mucus and my throat was starting to get all choke-y from straining for breath, but lying down for about 10 minutes made all the mucus flow and I am fine again. I find that lying down on my side (with the more clogged nostril toward the ceiling) probably works best, but one can experiment. One thing that's better with lying down (compared to saline spray) is that with saline spray, you blow your nose after doing the spray, which make even more of the deeper mucus come up (but not completely out), which sometimes clogs things up even more (at least temporarily). Lying down just pushes the mucus downward, so that you can swallow it, rather than moving the mucus to a critical narrow spot in the sinuses. I still don't know why this happens on some days but not most days (haven't found any patterns with temperature or humidity, so far). It almost always happens in the morning (where I can breathe while I'm lying down in bed, but shortly after getting up things start to clog up). Lying down also rests the neck/throat which helps it to relax.
- Turning on AC/playing around with humidity or temperature. There might be U-shaped response with humidity. Too wet like steam and that will make your mucus loose, but too dry will also make your mucus loose to try to compensate for the dry air? In the middle is where the mucus gets thick?
- Certain emotions, like embarrassment, seem to open up my nose. Also bowel movements. Maybe there's a way to recall an embarrassing memory to open up one's nose?
- Standing up and pacing back and forth (even without pinching the nose, as in Patrick McKeown's exercise) can sometimes open up the nostrils, for whatever reason. But doing this is not physically trivial for me given my post-viral condition.

11.2 The better your breathing, the better your nose

There is a positive feedback loop between your breathing and your nasal clarity. In other words, the better your nose feels, of course the better your breathing will be. But perhaps surprisingly, the better your breathing is, the better your nose will feel!

Important note: as you start breathing through your nose more, you should find that your nose becomes less blocked even without doing these things. I used to need the nose strips *a lot* when I first started nasal breathing. But now I hardly need them. So these techniques should be thought of as temporary crutches – or like training wheels on a bicycle. They are there to help jump start your nasal breathing until your nose is strong enough, but you shouldn't need them forever. I had a clear shift when I first switched to nasal breathing, and then a second shift when I switched to *silent* nasal breathing. It seems odd that my nose would clear up with *less* air moving through the nose. My hypothesis is that either my health in general got better, or maybe less air means less irritation. Lippmann (2015) mentions that “the thickness or thinness or real-time mucous production, as well as nasal turbinate size are regulated by CO2 levels”, so that's another possible thing that maybe happened to me.

Definitely don't get surgery, unless you've tried all the safe options and nothing works. I think the mainstream medical world understates the risks here, e.g., you should at bare minimum know about empty nose syndrome. Nestor (2020, chapter 7) has a bit about this. See also Sager (2024) for a random encouraging anecdote that I came across on Twitter.

my nose still sometimes feels blocked in the morning. not sure what's going on there, but i have a guess that the reason my one nostril plugs up during the night is actually functional. breathing needs are reduced during sleep because you're just lying there and using no muscles. so even the quiet gentle breathing style i have going during the day would be too *much* air. so my body figures out a way to reduce the air flow, which is to block out the one nostril. this is why my nostril is plugged up, but i seem to be breathing solely through the nose the whole night, even if things are blocked! *and* it's also why as soon as i wake up, the mucus clears up, because my body realizes “wait, now i'm awake and upright and active, and this is too *little* air now, we must unblock the nostril!” So for a few moments in the morning i feel a little uncomfortable and might have to take a few forceful nose breaths or even sometimes mouth breaths, but then

my body adjusts within 30 minutes to an hour and i'm ok again.

2024-10-16: over the last 3–4 days, i just stopped taking mouth breaths (i take maybe 4 mouth breaths per day now? whereas i had been slacking on my breathing and taking more like one every few minutes for the last few months, especially in the mornings/daytime when my breathing tends to be worse). and just doing this has basically fixed my nose plugging up in the mornings.

11.3 Mouth taping and chin strapping

(probably move this section to somewhere else)

i should say something about mouth taping as well. i don't have any *strong* argument against it, but i have personally not done it (except for briefly during the day). as you improve your breathing, your mouth should automatically close. so mouth taping is, in my view, unnecessary in the long-term. what about the short term? after all, all the other nose-unblocking stuff i'm recommending are short-term crutches as well. i think in the short term maybe it's completely fine? i was just personally scared of using it, and never needed it. waking up in the middle of the night gasping for air is pretty horrible (i've had it happen in the past even without mouth taping).

11.4 Picking and blowing your nose

i should eventually say something about nose cleaning – whether to pick, blow, etc. i tend to have a habit of blowing my nose when i feel too much watery mucus build up, and then i pat the insides with tissue paper. but this is probably not good. i've heard the inside lining of the nose is quite delicate and you should never pick or clean with a tissue. it's just kind of annoying to feel the sensation of the stuck mucus (how do you remove the blobs without picking them?) Even with a nasal spray i still have wet blobs sticking around. 2024-06-08 update: when i started doing relaxed buteyko style breathing every day, mucus production dropped by a lot and I didn't even get the urge to blow my nose. so i think the desirable long-run end-state that your breathing becomes so good that you basically don't have any mucus coming out of your nose ever, you don't even feel the urge to pick your nose, and your nose is just this cool self-cleaning organ.

Sometimes you *do* just have dried mucus lodged into a particular spot and it is clogging things up and if you just pick it out it feels *so* much

better. I'm not sure how to reconcile this personal experience with Buteyko theory(? or maybe just practitioners) that says you shouldn't blow or pick your nose.

Sometimes, blowing your nose *is* the right answer. e.g. on the morning of 2024-06-15, I felt the presence of both wetter mucus and dried up mucus deeper down, and I could feel it was blocking out my nostrils some. I decided to blow my nose, and the big chunk of mucus came out pretty easily, without blowing hard at all. And I felt *so* much better in my nose afterward. This has to be a sign that it was the right decision. Longterm goal is still the same, to breathe so lightly that your nose just doesn't produce excess mucus and it's self-cleaning so you never really have to do anything to your nose. But in the beginning stages, I think blowing your nose is still sometimes the right choice.

My nostrils often *do* feel better after blowing my nose. I think best time to do it is right when I wake up and flip over on my belly, I've noticed my mucus is looser in that moment. But even later can work. Blow once to clean out the mucus that built up during sleep, so your body can adjust to the breathing requirements of daytime. I don't do this every morning though. Still trying to figure out what's best.

11.5 Differential diagnosis ideas

In early May 2024, I did some house cleaning that created a lot of dust. My nose wasn't blocked, but there was a lot more resistance. And I noticed that nasal strips were helping, whereas NeilMed saline spray didn't. So in this case, I believe it was not *mucus* that's blocking the air, it's the actual turbinates. I think the allergies must have activated turbinates.

2024-05-07: this type of SOB feels *very* similar to the one I was getting all the time before the CapnoTrainer. So I wonder if allergic reactions can somehow like tank your CO₂. Or mess with your bicarbonate buffers? because the alternative is that I was exposing myself to allergens a *lot* back then for some reason, which just doesn't seem likely? since like, I wasn't dusting all the time. Or maybe I'm just having more turbinate activating, and that's leading to SOB feeling. And turbinates can be activated by both allergies *and* by overbreathing? (This last one was the theory I eventually settled on.)

11.6 What to do when you have a cold (or your body is otherwise producing a lot of mucus)

starting 2024-03-16 i've been taking nystatin to treat a candida overgrowth. this has caused a lot of classic die-off reactions in me, the most relevant of which here is a lot of mucus production in my sinuses. this makes it impossible for me to nasal breathe sometimes. so i've been thinking, "what can you do, if you get a cold or similar where your nose is just clogged up semi-acutely (i.e. over the course of days or a few weeks)?"

one response is something like "don't worry about it, anything you do in such a short span of time is meaningless compared to all the rest of your life". but this is unsatisfying since it's pretty unpleasant to not be able to breathe (and in my case, it was so bad for about ~3 nights that i had trouble sleeping as well).

some solutions i've used:

- neilmed spray – this is working well when the mucus is only kind of annoying. but at my peak, i would use it and then would breathe well for 3 minutes, then the mucus would come back.
- blowing my nose a lot – this was my main strategy, but i think i overdid it at times because i started seeing some blood mixed in my mucus.
- lying down on my sides instead of back (this was useful on the worst nights).
- learn to breathe through your mouth in the least disruptive way possible. rather than gulping or straining for air, just try to breathe softly, as if you are breathing through your nose.
- humidifier – this thing did not do anything for me.
- NAC – noticed slight loosening of mucus, but not enough to really help. i think i took as high as 1200mg(?) at once.

other ideas i haven't tried yet:

- raised pillow/slanted bed – i think this would have been useful on the nights when i had trouble sleeping.
- NyQuil/Tussin – cold medications/expectorants. I have heard from friends that these are quite helpful during colds.

11.7 Breathing trouble when trying to fall asleep

For the last several months or more, I've had this thing where I am breathing completely fine when upright, but then when I lie down to sleep, I am fine at first, but then as my nervous system adjusts to sleep, something happens where my body freaks out and feels like it can't breathe. Most nights, it feels like my brain has trouble handing off the breathing rhythm to the night time rhythm, and it feels like the tempo of breathing gets dysregulated. But other nights it feels like there is mucus and this clogs up my nostrils just enough that I have trouble breathing.

A particularly bad instance of the mucus version happened on the night of 2025-12-23. I had a "sensory whackamole" where I cycled through can't breathe → sitting up so I could breathe → feeling cold → lying down (under covers, so warmer) → mucus clogged, can't breathe → sitting back up → feeling bored → reading on phone → too tired to actually read → lying back down → mucus clogged → trying to breathe harder → chest and diaphragm hurt → sitting back up → etc. etc. etc. for like 3 hours.

Since then, I have discovered that on nights I am susceptible to this dynamic, lying down the first time for the night is okay, but if I get back up to e.g. go to the bathroom to urinate, then lying back down the second time causes the mucus problem to appear. Something to do with gravity shifting mucus around the sinuses, maybe. So I have a general rule now where I try not to get back up once I lie down for the night. I also find that lying down on my left side initially, then sleeping on my back, seems to help (I used to lie down on my back first, then shift to my sides).

As for the nervous system variant of this general problem of "can't breathe when trying to fall asleep", I haven't figured out as good of a solution. Turning off my computer ~2 hours before going to sleep, doing a Buteyko session ~1 hour before going to sleep, just generally trying to not do too much at night in the ~2 hour period before going to sleep, seem to all help. I think this symptom will improve as my nervous system health improves (from brain retraining/mindbody work).

12 Allergies

I've had an odd experience in August 2024 where I kept waking up with my nostrils feeling narrower and my throat feeling like it was choking (from straining for air). Eventually I discovered that sitting outside pretty

much immediately made all of this go away (at the time, I was/still am house-bound and mostly room-bound so it's actually not a trivial thing to just sit outside, hence why it took me so long to figure this out).

Washing my bedsheets alone did not seem to do much. Vacuuming the floor (I have a carpeted floor) actually seems to have maybe fixed the problem. (Due to my severe decline starting in July, the floor hadn't been vacuumed in probably over 2 months.)

Taking claritin also seems to have helped on the one day that I tried that.

One takeaway point here is that "allergies" can look not like very stereotypical allergies (of sneezing, itchy throat, etc.), although I did occasionally have those more classic symptoms as well.

One model I have is that allergens enter your nose when you breathe in, and then they get "stuck" inside the mucus and continue to cause problems. When this is the case, NeilMed saline spray seems to work really well; by adding moisture to the mucus and loosening it, one can blow out the allergen-ridden mucus.

13 Techniques vs habits

One of the disagreements I've seen in the breathing world is the emphasis between techniques vs habits. Those on the techniques side (such as anxiety specialists and Buteyko practitioners) present various techniques you can use when you are stressed, anxious, or have breathing difficulty. Those on the habits side (such as Peter Litchfield) say that techniques are mere bandage covering up a dysfunctional breathing habit, and that the solution is to unlearn the old habits and introduce new habits.

(actually i think *some* buteyko people have basically the same views as me.)

I find myself somewhere in between these two views. In a sense, I think both sides are right, and it's actually quite important to understand what's going on here. Techniques are a way of *learning* a thing. Once you've learned it, you don't need to "follow a technique" anymore, it's just a part of you. But having it verbally, explicitly, legibly, outlined is a useful step in actually learning the thing initially. You follow the technique, you get anxious about whether you're doing it right, you get obsessed about it, you get good at it, and then finally you get bored of it. It's all part of

the *overlearning* process. Once your bodymind digests it, it poops out the scaffolding of the technique and you can just Do The Thing. See also the four stages of competence (Wikipedia contributors 2023).

So yes, do the techniques. Learn them so well that you get bored of them and forget about them. Find the technique that you can do all day long, such that the technique *is* the habit.

13.1 Buteyko as coping mechanism (i.e. techniques) and Buteyko as habit

a new way i've been framing the above is that buteyko can be *used as* either a coping mechanism or as a habit. in other words, it's not really that a thing "is" either a coping mechanism or a habit, but rather, it can be *used as* either thing. it's the way you use the thing that determines it, not some inherent quality of the thing. sometimes my nose gets blocked by my turbinates (or whatever) and i use buteyko as a coping mechanism to unblock my nose. but most of the time, i try to breathe through my nose silently, and that's using buteyko as a habit. the eventual goal is to use it only has a habit; you have a new baseline and you don't run into breathing problems and can just live life. but in the short term, it's ok to also use buteyko as a coping mechanism to get yourself to that longterm state.

now of course, there are some things, like nasal strips, that are *just* a coping mechanism. but buteyko is one of those fluid things.

14 Capnometry biofeedback

[TODO: fill out this section. It should be about how to use the capnometer well. maybe include my list of "things to try on the capnometer". talk about what the ideal breathing curve should look like.]

Given how expensive capnometers are (and potentially difficult to access as well, if you don't happen to have anyone near you who can rent them out), you might think they are not worth it. I think it's fine to be skeptical of them. I made *some* substantial progress without a capnometer, but then 1–2 months after switching to nasal breathing I hit a plateau where I was still having significant shortness of breath while talking (and okay levels of shortness of breath while just sitting quietly). It was only after getting access to a capnometer that I had a second big shift in my breathing. See the section The unreasonable effectiveness of capnometers for more about

why I think they are useful. But also, definitely try fixing your breathing without a capnometer first, and just see how far you can get! (reminder that i don't sell or rent out capnometers, so i don't benefit either way from your decision.)

the other important thing about capnometers: taking Control Pauses is frankly scary to me, and *some* people also seem to get unpleasant side effects when they do it (even when they're doing it correctly – artour rakhimov talks the most about this, of the people i've read). so a capnometer is a way to measure progress in a completely safe way. (update (2024-02-25): i think i've warmed up to taking CP measurements a bit more now. the key is that if you're doing it correctly, then you let go of your nose on the moment of the first inkling that you need a breath, way way way before any discomfort. people have a tendency to hold on too long, as a way of impressing themselves or others, because they want to feel like they are healthy. if your CP is half a second, then so be it. the important thing is that over time, no matter how small your initial measurements are, they will improve. i do still think some people get weird negative symptoms even if they correctly implement the CP, and this probably includes me, so even though i'm warming up to the idea a bit more, i am not planning to start *daily* measurements of my CP, although i will probably check it once every few weeks maybe, just to see if it's still causing problems.)

TODO: what the heck do you do when your baseline CO₂ is normal according to the capnometer, but you still have that slight air hunger at all times (like, it's subtle enough that i wouldn't have even logged it in my symptom tracker before), and it's been over a week since you've been in this state (so bicarbonate levels should have adjusted by now)? that's the place i am in now (2024-01-29), and i'm not quite sure where to go from here. maybe trying to have this same breathing as i'm doing light exercise? or practice maintaining good breathing while i'm talking (my CO₂ goes down and i feel more out of breath when i talk, although it's gotten better as my baseline CO₂ has gone up)? or do i need to go into more advanced buteyko exercises like reduced breathing and breath holds? there was that thing in Chaitow, Bradley, and Chris Gilbert (2014, p. 6) Box 1.1 which says "General levels of deconditioning (the opposite of aerobic fitness) lead to altered forms of energy production (anaerobic glycolysis) that encourages acidosis – hence a greater tendency to hyperventilation" so it's possible i am also just deconditioned, and if i can just exercise more each day, everything will sort itself out. (as i write this i've only been going

on walks of like 10–35 minutes in length and some stretches throughout the day. i used to run more but that routine keeps getting disrupted.)

14.1 The unreasonable effectiveness of capnometers

One might naively expect that capnometers would be useless, since they don't actually *do* anything to your breathing. All of the breathing changes you make are ones you make yourself, and the capnometer just tells you what your CO₂ levels are without affecting them. One might thus conclude that capnometers are useless. I don't have a great reply to this – in fact, some people seem to get better just by doing Buteyko breathing “blind” (i.e., without seeing their breathing curve using a capnometer). So clearly it *is* possible to do the same things without a capnometer. (Even Peter Litchfield admits this to some extent.³) But in my case, it was only after I got a capnometer that I saw rapid progress. Here I will try to build a model of *why* having a capnometer is so useful.

My model of why capnometers are so “unreasonably effective” is basically that with Buteyko breathing, part of the problem is that even if you make progress, you can never know if you're pushing so hard that you've overshoot your CO₂ and are now having too much CO₂. You can't really avoid this problem because both too low CO₂ and too high CO₂ feel pretty much exactly the same – air hunger. So a capnometer is what actually lets you know for sure whether you are undershooting, just right, or overshooting.

It's just super helpful to *see* what your breath looks like. Buteyko breathing can feel *scary* because you don't know if you're going too hard and might be harming yourself. Hooking yourself up to a capnometer tells you whether your CO₂ levels are normal.

³From Litchfield and Reamer (2024):

You may be wondering, “can I benefit from the course without using technology?” The answer, of course, is a resounding YES. The ultimate objective of breathing behavior analysis work is NOT about learning to use technology for manipulating breathing with physical techniques, i.e., an outside-in paradigm, but rather it is about learning new breathing habits that automatically optimize respiration, that is, aligning breathing with respiration based on the principles of self-regulation, an inside-out paradigm. The CapnoTrainer GO technology is a powerful tool for making this transition from an outside-in to an inside-out paradigm, that is, for identifying existing habits and learning new ones.

one of the things i'm still trying to figure out is, supposedly Konstantin Buteyko originally had various machines to measure CO₂ or whatever, but then they were destroyed, so then he was forced to come up with the Control Pause as the sort of replacement metric (update(2024-02-10): actually, after reading the "destruction of the laboratory" book, i think this isn't quite right. buteyko already had CP as a tool when he still had access to the combine-complexator, which is why he could make charts showing how certain pCO₂ levels correspond to certain CP lengths). but what i'm wondering is, what did his machines measure? was it just end-tidal CO₂? and if so, then what did he do after people reached a normal 40 mmHg of pCO₂? was that the end of treatment? or did he try to push the CO₂ even higher? Update(2024-02-10): See Altukhov (2009). for some reason, the CO₂ measurements are super high, and so apparently he just kept going until people reached the high CPs and high CO₂s.

all of the above being said, one of my goals with this document is to be able to write down instructions that are so precise that i could hand it to my past self who has no access to a capnometer and still completely cure air hunger. there will be a lot more guesswork/"feeling around blind" going on, but if the instructions are good enough then they should still work.

cf. also Rice (2024b).

15 Breathing exercises

Reminder that none of these exercises will work if you can't breathe through your nose comfortably. See the section Clearing your nose for how to unblock your nose.

one of the things i've personally found frustrating is when people say "just do buteyko" or they link to a video with super vague instructions. so, at the risk of overfitting to my own experience, i'm going to give pretty precise instructions for exercises and aspirationally try to make them be difficult to misinterpret.

it's probably overwhelming to see all these exercises and giant walls of text. one of the nice things about working with a breathing coach is that they will guide you by only introducing one technique/exercise each week or so, so that at any given moment, you have *one* new thing to work with that you are trying to integrate into your practice. so a challenge for me is:

if the reader isn't going to be working with a coach (which is presumably why they are interested in this document), then how can i make the exercise sequence friendly to them? i really need some sort of timeline or flowchart that can be followed, so that as little thought as possible needs to go into making decisions. it's all laid out, you just follow a recipe.

unfortunately the theory given in the section The biological basis of shortness of breath only says *that* you should breathe less to build back the CO₂ levels over time, but not *how*. The exercises given here are much more an art than a science. I'm mostly just writing from personal experience about what worked best for me. The only way you will know for sure if you are raising your CO₂ levels is to get a capnometer and measure your own levels.

15.1 Main practice: relaxed silent nasal breathing

This one is most similar to Rakhimov (2013, section 4.6, "Relaxing the diaphragm (Buteyko relaxed breathing exercise)"), but Neil Tranter also has a similar video (Tranter 2023). Andrey Novozhilov (2012) has similar exercises too (e.g., Exercise 1). Some Buddhist meditation techniques are also very similar to this. In particular, the first 25 minutes or so of Taft (2020)'s "Just Shamatha" guided meditation is strikingly similar. I recommend reading/watching these other sources to get these other people's takes on the material; they are, in my opinion, all pointing to the same thing, gesturing at that thing in different words.

However, in my experience, all the other buteyko resources are pretty much "you just sit there and relax your body and breathe lighter", and people will just go like "ok. . . . so you just sit there and relax and breathe? how hard can that be?" And I think this style of pedagogy will tend to disappoint people when they find out how tricky the practice actually is. Or people will just sit there and try to relax and breathe, and they won't make any progress because they do the exercise wrong?

(Further rant on Buteyko pedagogy: some buteyko-oriented teachers don't emphasize relaxation enough. Buteyko breathing is more about relaxation than it is about breathing. You really do not want to restrict your breathing by tensing up more; that's just going to make the air hunger worse. The Russian-lineage buteyko practitioners tend to be better in this regard, as they make sure to tell you to relax as you reduce your breathing. However, the problem with buteyko pedagogy in general is that it's not

really tailored to people with air hunger or people with a hypersensitive nervous system (such as those resulting from a post-viral illness), so it just kind of assumes you can do the exercises or can take your CP, and even that you can “just breathe less”, or that you can pay attention to your breathing without massively messing things up. People with air hunger issues (including myself) can find it very hard to measure their CP. Notice that in this document, this main practice is to be done *after* you’re already nasal breathing, *after* you’ve gotten comfortable with short breath holds, etc. And this is considered the most gentle of the buteyko exercises, so if this was a buteyko document, it would be the first exercise that would be taught. When you have air hunger, the gentler the better.)

If you have trouble relaxing your exhale or knowing whether you are relaxing your exhale, try the Intermittent technique: Double inhale technique or try artificially slowing down just the first part of your exhale. These two techniques should give you an easier time relaxing the exhale so you will know what relaxation feels like.

note that i use the word “silent” rather than “quiet” because for a long time i thought my breathing was quiet, but it turned out my breathing was still making quite a bit of noise compared to what i *could* do. i don’t want others to make the same mistake i did. if your hearing is normal, you should not be able to hear your breathing. if you are hard of hearing, then maybe get someone else to check for you.

It’s okay to have mild air hunger throughout the exercise. But you don’t want too much. While your CO₂ tolerance is still very low, you might only be able to do the exercise for a short time, or you might need to do things like artificially lengthening the inhale.

an open question for me currently is: after you do this conscious relaxed silent breathing, you will build up some mild air hunger. now, *after* you complete the exercise, how long should you “keep” that air hunger going? should you feel free to just take a slightly deeper breath and “clear” the air hunger, or should you keep it going for some time (if so, how long?), or should you just try to keep it going as long as possible? i think trying to keep up the silent breathing with slight air hunger for long enough periods tends to cause a sudden burst of anxiety for me, like a sense of panic at not being able to breathe. so i think that’s probably not the right thing to do. but it’s confusing because i go from feeling totally comfortable with the slight air hunger to suddenly feeling panicky. Lippmann (2015) says “This process [raising CO₂ tolerance over time] is long and finicky and can

cause very unpleasant symptoms. Less is more; barely knowing whether you're doing anything is best. You can also trigger panic attacks or asthma attacks if you're susceptible." Another thought is to, whenever possible when taking deeper breaths, to do so in more Buteyko-aligned ways, e.g., through the nose and as gently as possible. So don't satisfy the urge for a deeper breath by inhaling with your mouth or yawning or being super forceful about it, just try to gently incline towards a deeper breath, a slight force with your nose. NOTE: this may not be possible at first, e.g., i had to get to silent nasal breathing in order to take "good" breaths with little force through my nose. when i was still mouth breathing some of the time, i always had to take a good breath through my mouth. then when i started nose breathing (but loudly), i could take good breaths occasionally through my nose, but only by loudly sniffing in. so there's a gradual "handoff" where the old normal becomes the new "rescue breath" method.

(2024-06-10: my current hypothesis that I am experimenting with is that you should basically never have to take a mouth breath/forceful inhale/straining inhale. If a reduced breathing session brings on so much air hunger that you feel like you're suffocating, then you went too far. And if eating or talking or walking or mentally or emotionally exerting is causing you to take a mouth breath, then you should stop whatever you are doing, and just calm down enough that you can breathe silently through your nose without any exertion again. If you really need to take mouth breaths because it's suffocating otherwise, then of course it's okay to take mouth breaths, but any mouth breath is a warning that you should just stop whatever you are doing and fix your breathing before going back to other stuff.)

Try slouching too as you do this exercise, and in general experiment with your posture. Your diaphragm will give you feedback on which postures work best. For example, when I slouch as I do this exercise, I notice my diaphragm can't move as well (both for inhale and exhale). Being too upright also makes it seem like my whole torso is too "stiff". So some intermediate posture seems to work best for me, but idk yet if this is optimal.

I think this exercise might be "breathing-complete" in the sense that it's the only thing you need to cure shortness of breath – but I am still actively testing out this hypothesis. i also suspect this exercise is "universal" in the sense that it is particularly easy to stumble onto. i sort of (with a bit of help) stumbled onto it while using the capnometer. the buteyko people also

recommend this (or slight variant) exercise for beginners.

relaxing completely is harder than it seems. I have a tendency to “relax” by like forcing my body to collapse, and that is not actually relaxation – on the capnometer i can actually see i am failing, because my CO₂ will go from like 38 to 34. you should not feel any strain or “tug” at all in your abdomen. i don’t know how to convey the actual precise instructions necessary to do this; i am just getting feedback from the capnometer. when the capnometer tells me my CO₂ just went down from the new thing i tried, i’m like “ok, i guess that’s not the right way to do it”, and go back to my baseline breathing style, and then try another thing.

for me, even on 2024-01-29, doing this exercise for 10 mins caused a mild suffocation-like reaction after about 5 mins, where i freaked out and stood up and had to take a few (unsuccessful/unsatisfying) mouth breaths, and then had to like just let my body do whatever for a few breaths, and then try to get in some good nasal breaths. after that, i was able to return to the exercise again, but i had to keep indulging good breaths. idk, this exercise is a lot harder than it seems! it’s weird though because i *feel* like i’m just doing this throughout the day, but i never get this suffocation reaction. so i wonder if it’s just that i’m indulging good breaths throughout the day, but during this exercise i don’t, and that’s the crucial difference? i think i should start lower, like do this for 2 mins without indulging, then next day go up to 3 mins, etc?

Litchfield and Reamer (2023, p. 57) has an interesting prompt, “Do you feel air hunger before the end of the exhale?” It made me pause because I always thought I continuously have air hunger, but actually, paying attention, I don’t have it on the inhale, even if it’s soft. Then as the exhale progresses it slowly mounts, in a subtle way. I’ve noticed that paying precise attention to when the air hunger comes one actually diminishes the air hunger.

eventually, i think the goal is to do this exercise while walking, then while jogging, biking, etc.

another point: you don’t want to just move your diaphragm. natural breathing actually involves movement of the whole torso, from upper chest to ribs to back to diaphragm. if you find that you are only moving your diaphragm, then this is a sign that one or both of two things are happening: (1) you’re exaggerating the movement of the diaphragm; and/or (2) you’re artificially suppressing movement of the rest of your body, tightening it up.

it might be helpful to do this exercise with a naked upper body and

staring at a mirror, so that you can see exactly which parts of your torso are moving in what ways (you can also try it in the shower but your neck will be bent if you want to look down at your torso). alternatively you could put your hands on various parts of your chest and feel the movement.

any reduction in breathing *must* come from relaxation, *not* from tension/control. this is a good video <https://iv.datura.network/watch?v=uvWSwxKUK1A> (that i totally did not appreciate in the beginning, because i wasn't even trying to do any reduced breathing. but what i've realized is that relaxation=reduced, at least if your breathing is awful like mine. and sometimes you consciously or unconsciously try to make your breathing more silent, and you really don't want to do that by tensing up.)

2024-06-03: this practice is so simple but so difficult. it can feel like there's barely anything to it, and you might deceive yourself that you are always doing it. i kind of had the mistaken impression that i'm "always doing relaxed silent nasal breathing". but then i actually sat down yesterday and listened to "you just landed on this planet" again, and kept going for like an hour (the recording itself is only 9 minutes long). i felt so relaxed afterwards, hands warm, salivating, and like my breathing cleared up. then today my diaphragm wasn't tensing up like normal, and i was able to talk for a lot longer! same thing happened this evening, my hands are warm (pulse oximeter recorded a perfusion index of 3.2%, when usually I am at like 0.5%), I'm salivating, feeling so relaxed that i'm kind of sleepy. this buteyko stuff is so real, it's crazy. but it's also difficult to get right – while listening to "you just landed on this planet", my core/chest/diaphragm area felt tight, and i couldn't get it to relax until well after the recording ended, by distracting myself by doing other than paying attention to my breathing. i keep shifting my posture around, and trying different sitting styles, to see which postures/sitting styles will lead to relaxation in which parts of my body. "belly soft like jelly" is hard. also, i am getting the thing again where even as i reduce my breathing, i can get "good" breaths in by like, "stretching" my insides somehow? normally i'd have to strain with my muscles or open my mouth to get such "good" breaths in, but i'm doing it almost accidentally, as the good breaths are "offered" to me. this also shows how it's ok to leave breathing stuff on the table for several months (i think last time i really focused much on it was like February? March?) and come back to it. it'll still be here, waiting.

also many buteyko places will tell you to do like 3 shorter sessions per day, but i am finding it easier to just do one longer one at night, like 1-2

hours before i want to go to sleep. i think it will help with going to sleep on time too. another benefit of doing it before bed is that you are more likely to carry good habits of breathing into your sleep (and hopefully have it still going when you wake up).

some sections that i want to fill in eventually:

15.1.1 Instructions for the main practice

Exercise instructions: sit upright in a comfortable way. Breathe silently through your nose. Then just try to make sure the exhale is done in a completely relaxed way, where your whole body (but esp diaphragm) relaxes. You are *not* pushing the air out. Just a whole-body relaxation on every exhale. On each exhale scan *one* body part for tenseness: shoulders? torso? legs? etc. Just focus on one body part at a time, and try to relax it on the exhale, and keep it relaxed. The idea of this exercise is that just by relaxing the exhale, your breathing should slow down (but don't try to artificially slow down your breathing) and so it actually *is* a (mild) reduced breathing exercise even though it just seems like a relaxation one. Do this for 15 minutes or so. As you're doing this exercise, you don't want to be forcefully inhaling, trying to take in "good" breaths – your inhales are silent, after all. (If your air hunger gets so extreme that you feel like you're suffocating, then of course it's okay to take a forceful emergency rescue breath, but this also means your silent breathing was too shallow, possibly artificially restricted, or maybe you are not ready for this exercise yet and need to do one of the intermediate/temporary exercises.)

Attention drifting during the exercise is a *good* thing because you want to make sure this style of breathing becomes a *habit* (rather than only occurring when you are consciously executing the instructions). You want your mind to drift away from your breath, as a way of giving yourself small opportunities to see if you can maintain the same breathing pattern even when you are not paying attention. The important point is that whenever you realize you have drifted and you're aware of your breathing again, you want to gently check to make sure you're still doing the completely relaxed exhale, and that your body isn't tense. You can even try to intentionally pay attention to something else, to distract yourself, so that you can check back on your breath to see what it has been doing.

The whole exercise is basically juggling your attention in this way: Is my inhale silent or as quiet as I can manage without tensing up? Is my

exhale completely relaxed? Are all of my body parts relaxed (except spine, to keep an upright posture)?

2024-07-18: i was lying down in bed, and felt like i was just exerting too much effort in my diaphragm, like my muscles felt kind of sore-ish, tired-ish. i was trying all kinds of things to see if i could relax it, like rolling to different sides, trying to make the breathing more light/shallow, placing a hand on my belly to guide it, trying to just will the muscles to relax. what finally worked was to make the inhales/exhales shorter. so instead of sucking in air slowly and releasing it slowly, each breath was like a short “bump”. i remember actually doing this on the capnotrainer too; it was how i got those nice sinusoidal waves, rather than the boxy breathing curves. making each breath shorter seems to make the breathing rate faster, but this seems fine. my muscles felt way more relaxed. it was also the first time in a while that my PI was up to like 1.8%, whereas usually it’s like 0.5%. it’s a bit counterintuitive because typically what i’ve seen (in things like box breathing, cadence breathing) is the recommendation to extend/lengthen then inhales and exhales. but my own experience here suggests that sometimes what the body needs is the opposite. (See also: note from 2024-06-17.)

15.1.2 Follow guided audio tracks while doing the main practice

it helps to have a guided meditation audio type thing to periodically listen along to as you do this practice. what I especially like doing is to start out by listening to a recording to prime myself, and then after the recording ends i just keep going on my own. i’ve been using the breathing normalization meditations from <https://buteykobreathingcenter.com/products/breathing-normalization-meditations-mp3> (my favorite recording is called “you just landed on this planet”; the whole collection is just \$4.95 and I wasn’t even charged any tax on it) but i don’t think there’s anything super special about these. visualization seems to help, e.g. imagining the air as having a pleasant but strong aroma (if you are not allergic to pollen, it might even work to go get some flowers or go out in nature and do the exercise there, although as of 2024-06-13 I have only ever done this exercise indoors), so i would try to find one that incorporates such visualizations. but different things might work for different people here. . .

Another recording I really like is Tranter (2024).

2024-06-13: I think one of the important things here is to have a good

guided audio track. I'd like to produce one at some point. One of the problems I am seeing with both my document here, and e.g. Artour's writings, Simon Spire's LNB Reference Guide, is that they are all written words, so it's hard to sync it up with realtime. Instead of saying "just loop through these instructions", it's nice to have that work offloaded to the audio track creator, and just be focusing on the present moment, having the instructions do the cognitive work of figuring out what to do when, so that you can just focus on fixing your breathing.

2024-06-22: doing buteyko without a recording today. It worked, and in fact it only took like 3–5ish minutes to reach the desired state. Hands are very toasty now. PI went from 3 to 6. So I don't think a recording is necessary, though helpful in the early stages. The problem is when you do it while falling asleep; I think that is where danger lies.

15.1.3 Signs you are doing the main practice correctly

according to the buteyko community, the main sign you did the main practice correctly is that you CP will be higher after the exercise, compared to right before the exercise. If you are one of the lucky people who can take CP measurements, then you can probably just skip the rest of this sub-section. In the rest of this sub-section, I will be assuming that you cannot take CP (e.g. you have some health condition that makes it bad to hold your breath, or holding your breath causes a suffocating/panicky sensation even when you are doing the CP measurements correctly).

Your hands may warm up a bit, and this seems like evidence that the exercise is working. However, hands can get warm even if you're not in the Good State. In fact, my hands tend to get warm out of anxiety/the slight panic that I'm feeling air hunger, sort of like how your face can get flushed when you are embarrassed. So in my experience at least, it's not the best sign.

other possible signs: feeling relaxed, sleepy, calmer, euphoric, extra saliva. I've also noticed that sometimes I start yawning repeatedly, and I'm not sure if that's a good thing or not yet.

2024-06-05: apparently i haven't mentioned this anywhere in this document yet, but i think perfusion index on a pulse oximeter (only some pulse oximeters have perfusion index as a measurement, so if you want to try this, make sure to buy one that does have it) gives a fairly good indication of how warm your hands are/how much bloodflow there is to your finger tips,

which is one of the ways you can tell whether the exercise is working. I like to take before and after measurements on the pulse oximeter. (2024-06-16: This matters way more if you (like me) can't or don't want to take Control Pause measurements. It's nice to have *some* way to measure progress/get some sign that the exercise worked, although with experience I think it becomes pretty obvious when it's working vs not? (I tend to feel a kind of euphoric calm contentment when I manage to do it right.))

15.1.4 The goal of the main practice is to enter the Good State

15.1.5 The goal post-main practice is to maintain the Good State for as long as possible

the goal is to eventually be breathing in this way all the time. so it's a "technique" but also it is going to become a habit.

A consistent observation from my own practice is that it takes a good amount of concentrated effort for me to first *enter* the Good State, but it doesn't seem to require almost any effort to stay in the good state once I am in it. I can just be sitting, I can be typing stuff, I can even go around and get tea from the kitchen or whatever. I haven't tried talking. But basically, you can just sit and calmly do simple activities, and that doesn't seem to cause any negative effects once you're in this state. In other words, you *don't* need to expend so much meditative concentration except for the first ~15 minutes (with practice, I think the time it takes to enter the Good State can be lowered). My hands are *still* warm, and it's been like an hour. Pulse ox says 97–98%, which is a good indication I am doing something right, because during the day it kept saying 99–100%.

15.1.6 The main practice is surprisingly tricky to do

15.1.7 Scheduled practice is necessary: the main practice is distinct from everyday silent nasal breathing

i think formally scheduled sessions (where you actually sit down with a timer and focus on just your breathing, while following audio or video instructions, and then recording in a spreadsheet) is actually very important. i think this might be why i didn't see any progress in my breathing symptoms during all of February 2024. it's very easy to convince yourself that you're "doing buteyko all the time". but if you actually sit down and

do a real session, you will notice your SpO2 readings go down (from like 99–100% to 97–98%) and your hands get warmer. and this is a sign that the session did something real, over and above what you thought you were already doing. my working theory right now is that it's easy to miss those slightly bigger breaths you occasionally take, and think that you didn't take them, or to be breathing with just a bit of sound rather than complete silence. but when you're doing a formal session, you're more careful and prevent those from happening.

i should note however that i don't regret avoiding formal sessions for so long – i have a preference for just doing what my body and mind want to do and not forcing myself to do anything. formal sessions seemed aversive because there's so much audio/video to choose from and no one ever explained why i should do formal sessions rather than just trying to change my habits. but at this point i think i'm still missing something and so i feel more motivated to try formal sessions.

15.1.8 Dangers with the main practice (what can go wrong)

Don't do it while trying to fall asleep. See Suffocation feeling for what can go wrong.

Don't try to go for a lot of air hunger. You only need/want a little bit.

15.1.9 Results of the main practice depend on both skill and initial state before the session

2024-10-12: starting at 7:35 p.m.-ish seemed too early to enter the good state. my hands are still cold after 15 minutes. but when i start around 9 p.m. or later it seems very easy to get my hands to be warm. during the day it's almost impossible to get my hands warm. i should try to figure out when the earliest possible time is when i can get my hands warm, and figure out what kinds of physiological processes tend to happen around that time (e.g. hormonal cycles). At 10:07 p.m. on the same day, my hands are warm and 98% O2 sat without even trying (whereas earlier i was struggling with air hunger at 99% and couldn't reliably get it down to 98%). so yeah, something definitely shifted in the last 2.5 hours. my throat is still chokey however.

15.1.10 It is possible to spontaneously enter the Good State

15.1.11 Tools to help with the main practice

Open question for me (2024-07-04): how useful is it to use things like neilmed saline spray and drinking hot liquid before/during the main practice, so that your hands get warmer (higher perfusion index)? I've noticed that if I do these things, it's much easier/faster to achieve a higher perfusion index and a lower SpO₂ (e.g. 97–98%, whereas usually I am 99–100%), and I do feel calmer and better. But does that translate into better breathing health? It probably doesn't hurt, but I might just be deceiving myself about feeling better. So I hope my breathing health will improve to the point where I can do control pauses without issues...

15.2 Introductory exercise: Many small breath holds

Tranter (2022) – the actual exercise starts around 7:46. also note that he counts kind of fast. i think his “5 second” count is actually more like 3.5–4 seconds (i should verify). 5 seconds is actually quite a long time to hold your breath, if you have air hunger! don't be discouraged if you can't do an actual 5 seconds. even tranter's count down is enough to get you used to the idea that deliberately holding your breath a bit isn't going to kill you.

i think the main value of this exercise is at the start, to get used to very small amounts of air hunger, to be like “ok, yeah, this is uncomfortable but it won't kill me”.

you can try doing this both when you're feeling mostly ok and when you're having more trouble breathing. for me it was very easy when i was feeling ok, but quite challenging when i was having trouble breathing (so much so that i'd have to take “rescue” mouth breaths every few breath holds). you don't want to follow the video so literally that you feel like you're suffocating; the idea again, is to just get used to doing breath holds and intentionally building up mild air hunger. so this is more of a *psychological* exercise than a physiological exercise.

i currently don't recommend doing longer breath holds. i think it helps for some people, but i think there is more risk, and i am pretty sure it's not required for fixing air hunger.

15.3 Intermittent technique: Double inhale technique

you take two relatively quick inhales, then exhale in a relaxed way.

This one was shown to me by my breathing coach. I've found this quite useful for getting my body to relax the exhale. i have no idea why it works, but somehow it's just easier to exhale in a relaxed way from a more full inhale.

15.4 Intermittent technique: Slowing the first part of the exhale

TODO: fill out. this one is similar in intent to the previous technique.

15.5 Intermittent technique: watching automatic breaths

Usually I have found that if I start paying attention to my breaths, my breaths become consciously controlled ones, i.e., non-automatic. This brings up a question of whether it is possible to “study” your own automatic breaths without controlling them (e.g., to find what your natural breathing rhythm is, or because your automatic breaths are better).

One natural idea is to just record yourself all the time, or to have someone watch you all the time, so that they can catch you when you're not focused on your breaths. But this requires a lot of setup and I am not even sure that your breaths will be the same when hooked up to a machine.

If you feel like even when you are trying to just let the breath naturally happen (when doing the main exercise) that you are somehow controlling or manipulating your breath (which expresses itself as discomfort/air hunger, pain in your diaphragm, a feeling that your throat is subtly being choked, etc.), then this exercise might help.

Exercise instructions: lightly tap your teeth kind of as if you are chattering from being cold. You can tap along to some song you like. What I've noticed is that since I have to tap along to the song, I can't control my breath as well, *but* I can still “watch” the breaths without controlling them. If you have more musical talent or experience then it may require a more complex pattern to distract you away enough from controlling your breath.

By tapping in a simple rhythm lightly with my teeth, like or something repeatedly, if I focus on the tapping, then my breathing becomes automatic. But because the teeth are close enough to the nose, you can kind of “watch” your nose automatically breathing.

One tricky thing is that when I tried this while walking, I found myself wanting to breathe at the same rhythm I was tapping with my teeth, which is not what we want.

I have found better luck when lying down on my back and just tapping with my teeth lightly, and watching my breaths.

15.6 Temporary exercise: soft smooth slow long inhales

TODO: explain this exercise. it's transitory because eventually (within like a week or two) you want to switch to the relaxed silent breathing. this exercise is a crutch for when relaxed silent breathing is too hard.

15.7 Temporary exercise: soft long exhales

this is the counterpart to the previous exercise.

i think the idea with these two exercises is that you're focused on just *one* half of the breath, and trying to do that half correctly. but the other half you can just do whatever. eventually, you will be able to just do the main exercise (i.e., both halves). in fact, you should eventually notice that slowing one half of the breath (esp inhale?) will lead to slowing the other half too.

once you start doing the main exercise, these transitory ones will feel extremely unnatural.

15.8 Intermittent exercise: observe your breathing after waking up

breathing during sleep is purely automatic. many people with air hunger report that their breathing is unusually okay during sleep and when they first wake up, but then becomes dysregulated soon after waking up. so the idea here is: as soon as you wake up, try to watch your breathing *without trying to change it*. how are you breathing when it's completely automatic? which muscles are you using to expand the chest area? are you "sucking in" the air mentally to a particular part of your body? what's the pace of breathing? how do you feel? do you feel air hunger?

it's ok if you can only do this for a few seconds before you take control of your breathing and it gets worse. each morning is a new chance.

"automatic" does not necessarily mean "perfect" – a lot of what we do automatically is carried over from our conscious breathing patterns.

(e.g. i've noticed that as I consciously altered my breathing to be better, it carried over into my unconscious/automatic breathing and that got better too.) BUT i think for a lot of people, their automatic breathing is sort of "one step ahead" of their conscious breathing. so the point of this exercise is to figure out what that next step is.

15.9 Intermittent technique: distraction

(2025-01-24: TODO: i've come to dislike the word "distraction" and now prefer "engaging". but the distinction here is tricky and it's easy to get confused and think one is engaging when in fact one is only distracting. so i will at some point need to come back here and explain the difference. to be honest, i am still figuring out the difference myself.)

when you feel out of breath (e.g., after talking) and just sitting there quietly isn't helping, this is a sign that you're still somehow, on some level, interfering with your breath and trying to control it. one way out of this, at least as a temporary measure,⁴ is to just totally distract yourself in some way so that you're not paying attention to your breathing at all. this will tend to "reset" your unconscious breathing pattern, back into a "good" state. you can try any of the following, or even anything else that you can think of:

- pace around the room (note: you probably don't want to go on a *walk*, which might make you exert too much and then you might feel a bit out of breath for a different reason, at least, if you happen to be deconditioned like I am)
- go make some tea
- go prepare some food to eat (even just getting leftovers out of the fridge and putting it on a plate and microwaving it has helped me on many occasions)
- go pee
- read something
- watch something, like a youtube video or movie

⁴Hopefully at some point I will figure out a better way out of such situations. I've found that like 80% of the time just sitting quietly and with decent posture and breathing silently through the nose will dissolve the air hunger feeling to a small enough level that I don't even think about it. So this distraction technique is helpful in the remaining 20% of times. But I'm hoping that I can figure out some better, less artificial, solution for what to do in those 20% of times.

- play a video game
- talk to someone (maybe using body language, sign language, gestures, or writing), hug someone, ask someone to hug you
- pet an animal (assuming you don't have allergies)
- take a nap
- brushing your teeth/flossing (personally I tend to notice my breathing is better after I brush my teeth and floss at night)

you can even combine any of the above, like pacing around *while* listening to a podcast.

you probably don't want to do something that requires a lot of mental exertion (like reading a difficult book) or that induces stress (like watching a thriller film). you want something that is distracting enough that you're not thinking about your breath, but also something that is "ordinary" enough that you're not unconsciously manipulating your breath in weird ways. a mild distraction, not an engrossing activity.

also possibly just for me, but eating/drinking is kind of stressful because i don't know how to coordinate the moments when i need to hold my breath when i swallow. so for me, eating food would not be a good distraction.

15.10 Intermittent exercise: timing how long it takes between big breaths

Background context: Over the summer of 2024, my general health had declined so much after tapering off nystatin that I wasn't really focused on my breathing at all. But my breathing actually seemed quite good; I wasn't getting air hunger most of the time. However, I did notice that my nose was plugging up a lot in the mornings, and so I would strain for air which would make my throat feel like it was choking me. And anytime I would get air hunger I would clear it with a forceful mouth breath. I was kind of deluding myself into thinking my CO₂ levels were fine because I was able to silently nose-breathe most of the time. But eventually I came to suspect that perhaps my breathing wasn't alright after all, that the morning mucus was there in the deep part of my sinuses *because* I was breathing too much at night. So this is the exercise I came up with on 2024-10-13 that really helped me over the hump, to get back into vigilantly reducing mouth breaths. By *measuring* and noting down each mouth breath, I had something concrete to work with, a metric I could optimize.

The exercise: sit with some kind of time-keeping device, and just try to breathe silently through your nose. Air hunger may build up. If the air hunger gets too strong, take in a big breath, then note down the time. Keep doing this until you can go about an hour without taking a big breath.

For example when I did this on 2024-10-13, I got the following result: 12:09, 12:13, 12:17, 12:26, (still ok as of 13:24). In terms of minutes since the previous big breath, that's 4, 4, 9, 58. By the end, I felt quite different.

What counts as a big breath? For me, any mouth breath, and any nose breath where I'm really striving for air. If I accidentally get a bit more air with a nose breath such that it becomes a "good" breath and clears the air hunger, that's ok. Yawning is ok (as long as it is a natural yawn). Burp is ok. A big breath is "third person" in the sense that someone sitting in front of you would be able to observe it and agree that you took a big breath.

I think it's ok to manipulate your breathing a little, not just observe your breathing. In other words, gently try to encourage your breathing to be lighter.

This exercise can be combined with the main practice, especially once you start being able to go for longer without taking in a big breath (so in my example above, starting around 12:36, say, I could have put on a recording to do the main practice).

The air hunger actually seems to dissolve after a while. So there's a point before which things are kind of tough, but then you get past that point and then you can just keep going forever (until some other event happens, like mental exertion or talking out loud).

You can even just buy a mechanical counter and count the number of mouth breaths you take in a day. As of this writing, they are sold on Amazon for like \$6.

First stage: If get urge to mouth breathe, stop everything, sit quietly, relax, and breathe through nose. If urge builds up too strongly, take mouth breath and write down the time.

After a week and urge to take mouth breath is no more frequent than 2–3 times per day, second stage: If get urge to breathe in such a way as to clear air hunger (breathe more forcefully, trying to get the air in deeper, etc.), then drop everything and just sit with the air hunger and relax. If air hunger too strong, then, *with the nose* clear the air hunger, then write down the time.

Stage three: equivalent to main practice/Buteyko-style reduced breathing. You can also instead do the same thing as stage 2, but do like a single

breath hold of like 3–5 seconds, to generate the *slightest* bit of air hunger, and then try to keep it going.

2025-01-24: a sensitized state (e.g. ME/CFS) + small stress (e.g. for me, even reading some paragraphs of text while I'm in AP) leads to breathing more to deal with the stress (because of prediction of increased physical activity; see the breathing disorders chaitow book). This leads to overbreathing, which leads to turbinate swelling, which leads to feeling like I can't breathe. And as we all know by now, trying to take bigger breaths while you feel like you can't breathe only keeps you stuck in that state longer.

All it takes to fix this is to write down the time, then keep writing the time when you take a big/forced breath. My nose opened right up after 10–13 mins, even though it was feeling closed up all day long until now. It's not a coincidence. This technique really works. I feel silly now that I came up with this exercise myself months ago, but only tried it at the end of the day. Somehow I was convinced this was caused by the AP I am in (and it is, if you consider just the "being sensitized" part; but there's more to breathing difficulty than simply being sensitized), even though I was breathing *better* in much worse states than this over the summer/fall!

Here's what I actually wrote down on paper:

9:23 [(pm). I only had to write down a single time!]

2025-02-04: Trying this again. I've tried it a few times since the previous update too, and it has worked every time. However, this morning's air hunger seems worse than usual.

11:30am

11:37am

11:39am

[nose is starting to clear at 11:42am]

11:44am

11:45am

11:49am [it may not seem like it based on the timestamps, but my nose is continuing to clear up]

11:50am

11:54am

11:58am – natural yawn

11:59am – another natural yawn

12:04pm

[and then I started checking my email at 12:12pm, which made me take a bunch more forced breaths as I tried to concentrate. . .]

15.11 Intermittent exercise: count how many times you blow your nose per day

16 Posture

I'm appreciating posture more now – it really makes silent relaxed breathing easier when you have better posture. But “good posture” is maybe not what society thinks is good posture, but whatever feels comfortable to you and allows for good breathing.

The Buteyko people also emphasize posture, but I have not read much about what they say. what i've seen is basically “sit at the edge of a hard surface with back straight”.

i still have a lot to learn here. apparently some posture techniques can be quite dangerous (can even cause die-off reactions?)

what can sometimes *feel* like air hunger might just be your abdomen being compressed due to slouching. sitting up a bit straighter can reduce the air hunger feeling. (this doesn't work most of the time)

i think posture matters more the lighter your breathing becomes. during reduced breathing exercises i think it's especially important. i notice if i'm slouched my breathing is more forceful. the problem is that i get back pain if i am sitting upright for too long (in supposedly “good” posture). so i need to figure what's optimal in the long run. there's got to be a way to get both good breathing and good posture and no back pain; i just don't know how yet.

i've been kind of going through a loop of {feeling tired → slouch → feeling like i can't breathe as well → sit up straight}. and then also sitting in different ways (cross-legged, seiza, normal, against back rest, not against back rest, which gives you like 6 combinations, and if you add slouch/straight then that's 12 configurations on the chair). and i think this is probably good, to just be changing posture and sitting style all the time, like every few minutes.

There are people with terrible posture who have normal breathing. So clearly posture isn't the full picture either.

question: why is the buteyko belt considered good, but slouched posture that restricts the diaphragm considered bad?

17 Breathing and sleep

One of the problems with going to sleep is that when you're lying in a dark room with literally no stimuli, it's very hard *not* to think about your breath and overly focus on it. And focusing on it too much leads to trying to control your breathing which often means worse breathing. This is a pattern I keep falling into, but I'm not sure how to get *out* of it. I mostly just try to get in bed only when I'm sleepy and also to think about random things while I'm trying to fall asleep so that I don't think about my breath.

18 Talking out loud

One of the types of shortness of breath I experience is tension in the diaphragm region that comes on if I talk too much. This particular type of shortness of breath has gotten worse and worse over the years, especially starting in 2023. I'm not sure what the physiological mechanism is. Here's a post I made on one of the air hunger support groups on Facebook:

Does anyone have tension or discomfort below the rib cage (diaphragm area) when talking, laughing, eating, etc., that makes it hard to breathe? I've been having this increasingly over the last several months to the point where I'm barely able to talk because I feel like I can't breathe. It's gotten so bad that I am sometimes passing paper notes to people rather than talking. Strangely though, if I stop talking I can just silently breathe through my nose, so it's like my breathing is "completely fine" in some sense, but also in another sense I still feel like I can't breathe, and also the tension/discomfort in the diaphragm area feels uncomfortable too. I've been having trouble putting my symptoms into words to find similar cases. Any ideas on what this is and what I can do?

I've had multiple chest x-rays, echocardiogram, pulmonary function test, methacholine challenge test, exhaled nitric oxide test, laryngoscopy, and two modified barium swallow studies. They all come back essentially normal with no clues as to what's going on with my breathing. My SpO₂ is consistently 97–100%. The one thing that was abnormal was my end-tidal CO₂, which was too low, but capnometry biofeedback brought that up to

normal. I've been doing Buteyko breathing pretty regularly and it helps to relax the diaphragm area temporarily, but if I start talking then it tenses up again.

So the optimistic line of reasoning is: because I've noticed a pattern where if I push over my talking limit multiple times in a day, each "pushover" feels worse and comes with less and less warning/comes with less and less talking, *maybe this pattern can be reversed, too*. If I can go, say, a full week without taking *any* forceful breaths or mouth breaths, will that "charge up my diaphragm battery" to "full capacity", such that I'll be like a normal person? Or does it only go up so far, and that amount is still way less than a normal person's (so that even at full capacity, I can only talk for 30 minutes or so now)? As of 2024-07-22, my experience has been that just resting my diaphragm/"talking muscles" by not talking does seem to restore some function – I've been able to say a few sentences at a time multiple times a day now with only slight tension (whereas in the past there were days where I couldn't even say a few words without feeling like my diaphragm region was going crazy with tension). I still need to type a lot to communicate though.

Also interestingly, it's not the kind of muscle soreness like I would in the past get in my arms or legs if I used a muscle a lot that I hadn't used in a while. And there's a lot of intra-day variability too – in the mornings I get way more tense, whereas in the evenings I can talk more. So I think it's actually not muscle weakness or soreness (if it was, I wouldn't expect things to get better so quickly), but probably more to do with the nervous system and needing to relax the muscles properly.

The advice i've heard about breathing while talking in the buteyko sphere/in the chaitow book are all about how to coordinate your breathing while talking, so as not to overbreathe. but this feels different from what i'm experiencing, which is this muscular tension/soreness feeling.

Some of the time I get some tension lower, in my gut area, from bloating. But most of the time it's a bit higher up, and not from bloating.

Update from 2024-12-09: I increasingly suspect this really is just from a hypersensitive nervous system issue, and is almost completely separate from the breathing issue. I'm currently working on the nervous system issue using brain retraining. I'm hoping this will take care of my other numerous symptoms too. This nervous system work will turn into probably a separate document, or a series of blog posts, or something like that. But

my first priority is to make sure this brain retraining stuff actually works.

19 Muscle tension

(Will have to write a lot more here as I figure things out. So far I haven't figured out very much...)

A hypothesis I have is that chewing, breathing, talking, swallowing, all make use of a lot of overlapping muscles, and they use these same muscles in *different* ways, so if you try to do multiple thing at once (like breathing while chewing), then your body can get confused, or like, it can lead to suboptimal patterns of both?

2025-04-06: ok, so i think the thing that feels off about the alexander technique people (e.g. simon spire) and the muscle tension people (e.g. the postural restoration institute) people is that they just kind of assume that you can intentionally let go of tension. like, they assume you can JUST stop trying to strain for deep breaths. and partly this is what is going on for people who overbreathe, and for some it may be enough. but you really need to ask "where is the tension coming from?" From a post-viral nervous system dysregulation perspective (which is where i'm coming from), the root cause is the nervous system stuff, and the nervous system activation is what is *causing* the straining for breath (brain thinks it is in danger, need to fight or flee), which leads to muscle tension. letting go of that tension isn't going to magically fix things; it will work until the next time your nervous system gets activated (from doing mundane things like talking out loud or programming or even reading, in my case). it's the nervous system that needs to be worked on, to get it to perceive less danger (dan buglio's term) in these mundane activities. and so this is where brain retraining comes in.

19.1 Greedy vs non-greedy relaxation

When intentionally trying to relax the muscles (e.g., using Buteyko), sometimes one gets stuck in local optima of relaxation. e.g. poor posture and gradual/incremental relaxation on exhailes will eventually produce a "cramped" feeling in the gut. The solution seems to be to *sometimes* undo relaxation, to find a better global state. Progressive muscle relaxation (PMR) works by a similar principle: that by *tensing up* one's muscles deliberately, one may be able to "find the knobs" by which to relax better.

20 Breathing synthesis

My synthesis = “Buteyko method” + “Alexander Technique” + Capnometry biofeedback

or maybe:

My synthesis = Nasal breathing + Silent relaxed breathing + Posture + Habits + Capnometry biofeedback + Brain retraining (anxiety and nervous system)

- Buteyko provides nasal breathing, silent breathing and a theory of air hunger via CO₂ levels. I put it in quotes because this is probably a vast oversimplification of the Buteyko method – but I only needed this much of the theory.
- “Alexander Technique” is what I mean by just allowing the breath to happen: smooth, not jerky. “Allowing the breath to happen” rather than doing the breath. And also making the good breathing a *habit* rather than “executing techniques” when you need them. Make it automatic/autonomic/subconscious/unconscious, rather than conscious/manual – this one might be two separate things actually (the smooth part, and the unconscious/habit part). Some of the Buteyko people also *mention* this, but it tends to be much more buried and not *emphasized*. I think it really needs to be thought of as one of the main components.
- Capnometry biofeedback is discussed elsewhere in this document, e.g., see the section The unreasonable effectiveness of capnometers. In some sense this one is *also* part of the Buteyko method, except that the Buteyko community moved away from capnometry after their equipment was destroyed/confiscated. (See Rice (2024b).) So in some sense I am picking and choosing the specific parts of Buteyko that I like, so it’s turning out to be less a synthesis and more a distillation.

21 Misconceptions

this section is mostly an index to stuff discussed elsewhere in the document.
(might remove this section if it doesn’t seem useful)

- CO₂ is bad – see Misconceptions about CO₂
- deep breaths are good

- diaphragmatic breathing is good
- you want to slow down your breath/breathe to a certain rhythm

22 Cheat sheet: emergency things to do if you're really having trouble breathing

(note: this section won't really be useful if you have not read the rest of this document already. it really is just a "cheat sheet" to jog your memory and give you a concrete list of prompts to run through, *not* intended as a magical cure for breathing distress. if you really are struggling and it's a lot worse than usual, then please seek medical help!)

things to ask yourself, or rather, here are some things i tend to ask myself, in no particular order, whenever i'm having more-than-my-usual breathing difficulty:

- can i distract myself using one of the suggestions listed in Intermittent technique: distraction?
- is the breathing difficulty coming from my nostrils being clogged or my turbinates swelling? do i feel resistance in my nose as i breathe (especially on the inhale, in my experience)? if so, can i use a nasal rinse or one of the other strategies listed in Clearing your nose?
- am i exhaling enough air? am i trying to just take in the next breathe before exhaling the current breath? can I exhale just a little bit more? breathing feels so much better when you are able to relax the exhale to its natural completion, and only then begin inhaling. When feeling air hunger, unintuitively it sometimes feels better to keep breathing *out* more air, rather than breathing in.
- Can I let the exhale end, and just let my natural breathing reflex start the next inhale?
- can i soften my inhale? can i make it silent, even just for a few breaths?
- does my brain feel like it's suffocating? if so, i can try to let myself take an occasional mouth breath.
- how does it feel to sit upright, place one hand on my upper chest and one hand on my belly, and to just tactilely feel my breathing? (I suspect this one has a slight calming effect from some deep part of you *knowing* that you are still breathing because you can feel that you breathing muscles are causing movement.)

- which part of my body is moving as i breathe? can i try to make my body move less, or distribute the movement more evenly across my body, so that i'm not just exaggerating movement in any one part of my body?
- am i slouched? can i be more upright so that the whole length of my torso is free to expand?
- can i change my posture? can i stand up? if i'm lying down, can i lie on my side vs back vs belly?
- can i check my oxygen reading using a pulse oximeter? is it 99–100% (indicating overbreathing)?
- Am I talking too much, or exerting myself (physically, mentally, emotionally) too much in some way? If so, I can just try to drop everything, and sit or stand or lie down and close my eyes and just not do anything for a while.
- when do i feel the air hunger? is it all the time? is it only on the exhale? is it at the end of the exhale? (peter litchfield's question.) Is it at the start of the inhale? is it during the inhale? (The latter two for me seem to happen when my body is confused and doesn't know how to breathe.) This one tends to help me pay attention to subtle qualities of my breathing, not just being like "I CAN'T BREATHE!!!"
- can i listen to one of my favorite relaxation guided audio recordings? (if you don't have one, make it a priority and find some to have on hand.)
- can i get "fresher" air (even if imaginary) by going inside or outside or to a different room or by moving away from a "pollutant" source? is there a smell that's bothering me? a lot of times, if you also have anxiety about your breathing, it is helpful to do things that might just seem like placebo, just to help your brain calm down so that it doesn't keep looping on and on about "what if i'm breathing in polluted air" or whatever.
- is my breathing *pattern* irregular? (e.g., pauses, or breathing fast for a few breaths and then slow, or differently sized breaths, like audible for a few breaths, then silent for a few breaths.) can i try to make it more regular? or does it really want to be irregular?
- does it feel more like i'm forgetting how to breathe (and so i should gently try to guide my body into breathing better), or like i am controlling my breathing too much (and i should instead just let my breathing happen on its own)? (i've felt both at different times and

it's quite confusing. i suspect the first one is some sort of autonomic dysfunction thing that i have because of my generic illness, and the second one is a more typical air hunger thing that i have due to overbreathing. but really, i don't know.)

- Can I relax some part of my body, or (if some parts of my body are already relaxed) *more* parts of my body? Especially on the exhale.
- is my heart rate also high? (for me, this tends to indicate i am having POTS and so lying down often helps me calm down a lot.)
- Can I take a nap or even just lie down and close my eyes? (This can feel scary and your brain may shout "No!" at the idea, but other times it's exactly what you need.) Sometimes it specifically needs to be lying down in a certain position, whether that's on your back, on one of your sides, or on your belly. On 2024-08-13 I had to lie down on my belly in order to relax my diaphragm; the other positions didn't help.
- Place a hand on my belly and just breathe at a regular pace. I find that saying in my head "1, 2, 1, 2..." to be better than saying "in, out, in out..." because when the air goes in, the belly goes out, which is confusing for my brain. "1" and "2" are neutral labels. Better yet, you can count down from 100 as you breathe in and out (I first learned the "count down from 100" trick in Neil Tranter's video <https://youtu.be/9EFnc3k-BCI>), which should also help to remove your focus so much from your breathing/anxiety.

Acknowledgments

i've benefited from interacting with a lot of people, in the course of figuring out my breathing. (people who i haven't checked with to see if they want to be mentioned are listed as initials, unless i'm only relying on public sources and no private interactions, which applies to Howard Cooper.) Being acknowledged does not mean that the person agrees with anything I wrote in the document.

- ML: kept mentioning buteyko breathing to me saying i should try it.
- Howard Cooper: While I've benefitted from all the sources I'm citing in the References, I want to call out especially Cooper (2021) as the thing that finally made the CO₂ stuff click for me (though my

understanding was still very incomplete) and got me to actually start playing around with buteyko techniques.

- CM: told me about the capnotrainer and got me excited about capnometry biofeedback.
- Vipul Naik helped me understand some key things about the biology of respiration. In particular, he was the one who figured out that having a low CO₂ means pH is more sensitive to slight fluctuations in CO₂ levels, just purely by the logic of acid–base chemistry. Also for reading drafts of this document and giving me feedback. Most of the things in this document I wrote up while discussing them with Vipul.
- DD: introduced me to JA. he was also a catalyst for me figuring out my circadian rhythm, but that’s a whole separate topic.
- JA: my former breathing coach; helped me get set up on the capnotrainer, and helped me through the whole biofeedback process.
- PL: was generous enough to spend an hour talking to me about capnometer stuff and responding to an email.
- KC: told me about the L1 acupressure point technique.

References

- Altukhov, Sergey (2009). *Buteyko Breathing Charts*. URL: <https://thebreathingman.com/buteyko-charts/>.
- Chaitow, Leon, Dinah Bradley, and Chris Gilbert (2014). “What are breathing pattern disorders?” In: *Recognizing and Treating Breathing Disorders: a multidisciplinary approach*. Ed. by Leon Chaitow, Dinah Bradley, and Christopher Gilbert. 2nd. Churchill Livingstone. Chap. 1, pp. 1–10.
- Chaitow, Leon, Dinah Bradley, and Christopher Gilbert (2002). “Self-help approaches”. In: *Multidisciplinary Approaches to Breathing Pattern Disorders*. Ed. by Leon Chaitow, Dinah Bradley, and Christopher Gilbert. 1st. Churchill Livingstone. Chap. 10, pp. 241–272.
- Cooper, Howard (2021). *Help with air hunger – I have anxiety and feel like I can’t breathe!* URL: <https://www.youtube.com/watch?v=XliOGg8Tl98>.
- Gilbert, Chris (2014). “Biochemical aspects of breathing”. In: *Recognizing and Treating Breathing Disorders: a multidisciplinary approach*. Ed. by Leon Chaitow, Dinah Bradley, and Christopher Gilbert. 2nd. Churchill Livingstone. Chap. 4, pp. 61–77.

- Lippmann, Mark D. (2015). *deep breathing can be bad for you, viz. buteyko breathing, having to pee all the time, sleep apnea, etc.* URL: <https://meditationstuff.wordpress.com/2015/11/05/deep-breathing-can-be-bad-for-you-viz-buteyko-breathing-having-to-pee-all-the-time-sleep-apnea-etc/>.
- Litchfield, Peter M. and Sandra Reamer (2023). *CapnoLearning: An Introductory Guide*. Breathing Science, Inc. ISBN: 979-8-218-16278-8.
- (2024). *CapnoLearning Course*. This citation is for the course homepage, not the contents of the course itself; I have not taken the course. URL: <https://www.thebsj.org/capnolearningcourse>.
- Nestor, James (2020). *Breath: The New Science of a Lost Art*. Riverhead Books.
- Novozhilov, Andrey (2012). *Buteyko Breathing Manual: Holistic & Drug-Free Method for Health Improvement*. Ed. by Tusha Yakovleva. 6th. Illustrated by Victor Lunn-Rockliffe. Breathing Center.
- Rakhimov, Artour (2013). *Advanced Buteyko Breathing Exercises*. CreateSpace Independent Publishing Platform.
- Rice, Issa (2023). *Next steps after modified barium swallow, endoscopy, swallow exercises?* URL: https://www.reddit.com/r/dysphagia/comments/1743nii/next_steps_after_modified_barium_swallow/.
- (2024a). *Analysis of Howard Cooper air hunger videos*. URL: <https://issarice.com/analysis-of-howard-cooper-air-hunger-videos>.
- (2024b). *Buteyko community and learned helplessness*. URL: <https://issarice.com/buteyko-community-and-learned-helplessness>.
- Sager, Sam (2024). *"quick crazy personal health story [...]"* URL: https://twitter.com/sc_sager/status/1749861513605558775.
- salamanderz1 (2023). *Does anyone else wake up gasping for air sometimes as if your body forgets that it needs to breathe?* URL: https://www.reddit.com/r/sleep/comments/10dvt0w/does_anyone_else_wake_up_gasping_for_air/.
- Taft, Michael W. (2020). *Just Shamatha*. URL: <https://deconstructingyourself.com/just-shamatha.html>.
- Tranter, Neil (2022). *BRAIN FOG – How I Cured My Brainfog by Holding My Breath*. URL: <https://www.youtube.com/watch?v=wjZL01lp5Ac>.
- (2023). *Buteyko Breathing Exercises – Relaxed Breathing for Calm*. URL: <https://www.youtube.com/watch?v=SEiXqQxq9mw>.
- (2024). *The Key to Buteyko: Learn the #1 Skill for Deep Relaxation*. URL: https://www.youtube.com/watch?v=aAlqZG9b5_Q.

- Wikipedia contributors (2023). *Four stages of competence*. URL: https://en.wikipedia.org/wiki/Four_stages_of_competence (visited on 01/27/2024).
- Yakovleva, Sasha, K.P. Buteyko, and A.E. Novozhilov (2016). *Breathe to Heal: Break Free from Asthma*. Breathing Center.